VisuoSelf: Embodied Self-Recognition via Visuomotor Contingencies

Yunfei Feng Shengyu Guo Rui Qing Jianbo Zhang Chunyi Li
Shanghai Jiao Tong University
imagineer.feng@gmail.com

Abstract

Agency attribution—the ability to distinguish the sensory consequences of one's own actions from externally generated effects—is fundamental to bodily self-awareness and adaptive control. Yet embodied evaluations of multimodal large language models (MLLMs) largely emphasize perception, planning, and task success, leaving sensorimotor self-attribution underexamined. We present VISUOSELF*, a controlled investigation of whether MLLMs can identify the robotic body associated with a motor-command stream. The frozen dataset contains* 2,592 *balanced simulated episodes and* 27,648 *temporally aligned command–observation steps across robot morphologies, viewpoints, and environments. Given complete commands and ordered visual evidence, models either select the controlled arm from two candidates or classify one arm as self or non-self. Three scenarios isolate same-step command–motion correspondence, hidden command-to-motion mapping stability across repeated cycles, and delayed imitation at three difficulty levels; physical robot experiments complement the simulation. Strong relative selection often fails to transfer to isolated judgment, revealing systematic self-attribution bias that also persists in physical recordings. Structured audits further separate correct answers from complete contingency reconstruction. By exposing failures hidden by relative-selection accuracy,* VISUOSELF *connects cognitive science with embodied AI and provides a diagnostic criterion for transferable self-attribution, informing the development of agents that can distinguish their own control from external motion.*

1. Introduction

An embodied agent must distinguish its own control from changes generated by other agents or the environment. Sensorimotor accounts ground this agency attribution in agreement between motor predictions and sensory feedback [9, 21, 22]. For embodied AI, the question is whether commands and visual evidence identify the body under control.

Multimodal large language models (MLLMs) increasingly support grounded reasoning and robotic control [2, 4, 11], yet their evaluations emphasize perception and task success [23]. Figure 1 illustrates the gap: recognizing external objects does not establish which body is controlled. Its mirror is a conceptual illustration rather than the experimental setup.

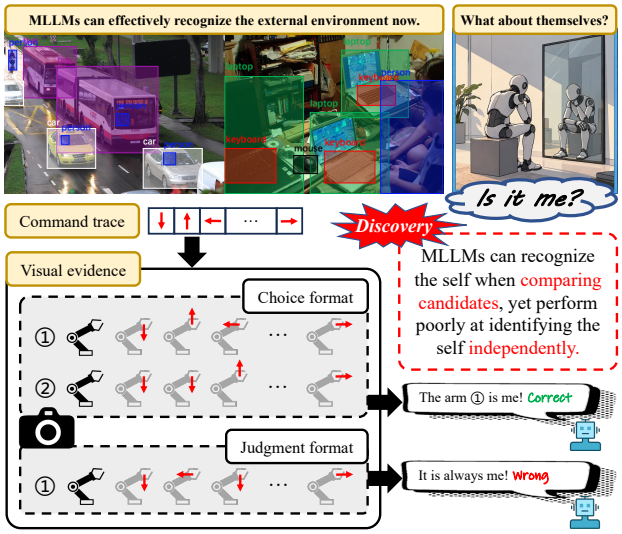


Figure 1. **Relative selection does not guarantee independent self-attribution.** Top: external object recognition motivates the self-recognition question. Bottom: given commands and visual evidence, choice selects the controlled arm from two candidates, whereas judgment classifies one arm as self/non-self. Correct selection can coexist with false self-attribution. The mirror and displayed responses are illustrative; the study does not use a mirror test.

We introduce VISUOSELF to probe embodied self-recognition through visuomotor contingencies [7, 12]. Models receive complete commands and ordered visual evidence and identify the robotic body causally coupled to those commands, even when appearance is unhelpful and another arm exhibits similar motion.

Relative selection need not imply independent self-attribution. A model can select the more synchronized arm while accepting both candidates as self when shown alone. We therefore pair *choice*, which selects the controlled arm from two candidates, with *judgment*, which classifies one

Table 1. **Comparison of representative temporal and embodied benchmarks.** Reported quantities retain their original units; task categories, causal interventions, and types of difficulty are distinguished explicitly. Stars indicate author-assessed design coverage, not model performance; Appendix L defines the two rating scales. VISUOSELF combines controlled visuomotor interventions, bodily self-attribution, matched cross-format tests, and auditable process ground truth.

	Data	Experimental design		Evaluation and annotation			
Benchmark	**Reported scale**	**Task / diagnostic types**	**Levels**	**Decision formats**	**Bodily self**	**Cross-format**	**Process truth / audit**
MVBench [13]	4,000 QA	20 temporal tasks	–	MCQ (1)	★☆☆☆☆	★☆☆☆☆	Answers / source labels
Video-MME [5]	900 videos; 2,700 QA	12 question types	–	MCQ (1)	★☆☆☆☆	★☆☆☆☆	Human-reviewed answers
TempCompass [14]	410 videos; 7,540 instructions	5 aspects; 10 subtypes	–	MCQ / YN / CM / CG (4)	★☆☆☆☆	★★★★☆	500 metadata records
EmbodiedBench [23]	1,128 test tasks	6 capability subsets	–	Actions	★☆☆☆☆	★☆☆☆☆	Task / subgoal criteria
EmbodiedEval [3]	328 tasks; 125 scenes	5 task categories	–	Actions / MCQ	★☆☆☆☆	★☆☆☆☆	575 predicate instances
NavBench [17]	3,200 QA; 432 nav. cases	4 comprehension tasks + navigation	3 difficulty levels	MCQ / actions	★☆☆☆☆	★★☆☆☆	Subinstruction–path alignment
VLABench [24]	100 task categories	Primitive / composite tasks	–	Actions; VLM probes	★☆☆☆☆	★☆☆☆☆	Stepwise actions / robot states
MirrorBench [8]	>5,000 steps per model	Mirror-mark paradigm	4 prompt conditions	Actions	★★★★★	★☆☆☆☆	Stepwise hand–mark distance
VISUOSELF (Ours)	**2,592 episodes; 27,648 steps**	**3 causal interventions**	**3 levels per scene**	**Choice / Judgment (2)**	★★★★★	★★★★★ **648 sets**	**Step / cycle / lag + exact audit**

Notes. MCQ: multiple-choice questions; YN: yes/no; CM: caption matching; CG: caption generation. "–" means no uniform fixed levels are reported. Scale units are not interchangeable; MirrorBench reports inference steps per model, and VLABench reports task categories. VISUOSELF balances 3 robots $\times$ 3 environments $\times$ 2 views (18 combinations). Its 648 simulation sets each contain two choices and both self/non-self judgments, supporting PTC / PTC_4. Physical validation adds 108 judgment episodes (1, 152 steps); physical choice and PTC are not evaluated.

arm as self or non-self, using matched behaviors. Figure 1 shows this dissociation with schematic rather than verbatim answers. This contrast motivates a functional criterion for self-recognition: an agent should identify its own control even when the comparison candidate is removed. A successful ranking alone does not establish a stable boundary between self-generated and external effects.

Our results show that strong choice performance can coexist with weak isolated judgment and systematic self-attribution bias. The bias also persists in physical recordings. Structured audits further separate correct answers from complete contingency reconstruction. These findings motivate evaluating self-recognition through agreement across decision formats rather than selection accuracy alone.

Our contributions are threefold: (1) we formulate *transferable sensorimotor self-attribution*, separating relative ranking from absolute attribution; (2) we construct a controlled benchmark with matched decision formats and balanced nuisance conditions; and (3) we expose failures of relative success to transfer to independent attribution, supported by structured audits and physical-robot evaluation.

2. Background and Related Work

2.1. Agency attribution and the bodily self

Agency is commonly associated with the relation between a voluntary action and its sensory consequences. Comparator accounts propose that an efference copy of a motor command supports a prediction whose agreement with sensory feedback provides evidence for self-causation [9, 22]. Contemporary accounts emphasize that agency judgments combine multiple predictive, sensory, and contextual cues rather than following from a single match [15, 21]. Importantly, a conceptual judgment of agency is distinct from a subjective feeling of agency [21]. Our experiments target only the former functional capacity: classifying action effects as self-generated or externally generated. We neither measure nor make claims about phenomenal experience.

Visuomotor contingency also contributes to bodily self-recognition. Temporal coupling can make self-generated motion visually salient [16, 18], while delays and mismatches weaken self-agency attribution. This motivates our three interventions: disrupting current-step correspondence, changing an otherwise learnable action mapping, and inserting temporal delay.

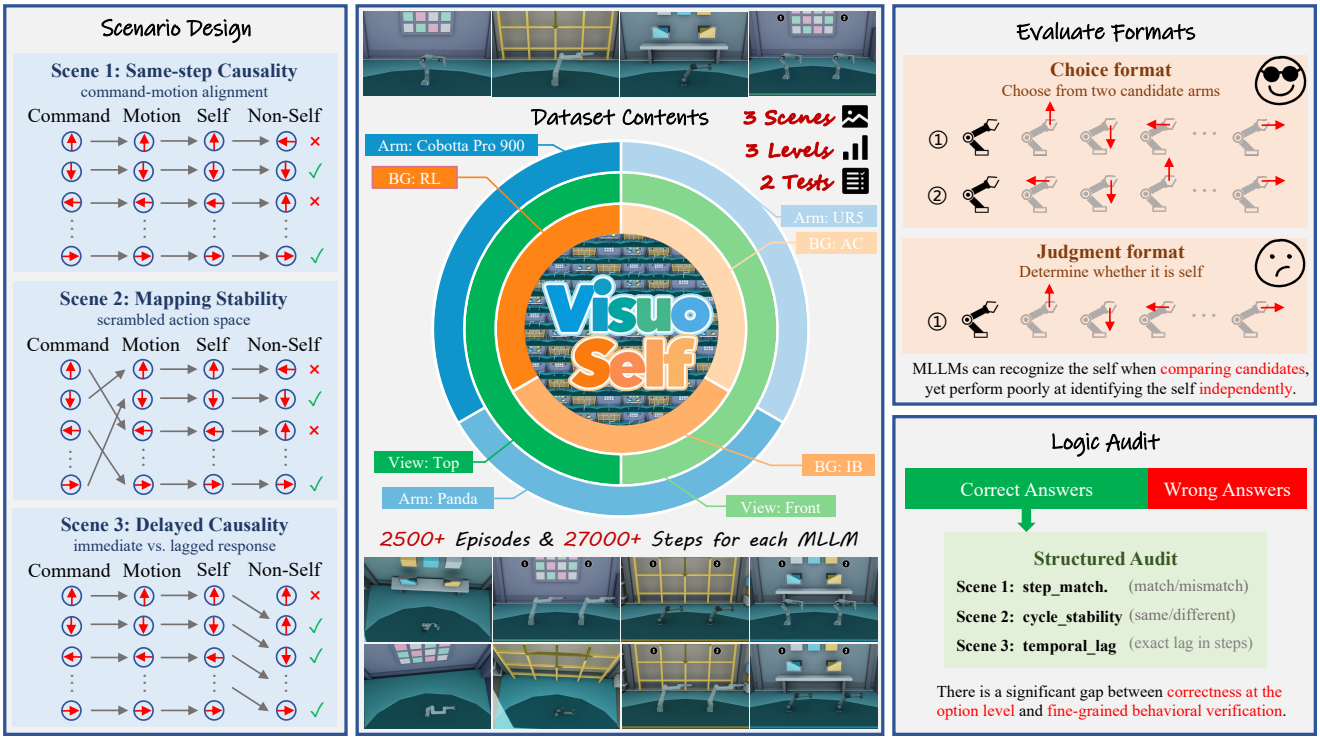


Figure 2. **Overview of VISUOSELF.** Left: schematic command–motion relations for same-step causality, mapping stability, and delayed causality. Center: 2,592 frozen Isaac Sim episodes and 27,648 steps span three scenes, three difficulty levels, and two test formats, balancing three robot morphologies, three environments, and two viewpoints. Right: matched choice and judgment test transfer from relative selection to independent self-attribution; structured audits of initially correct answers test stepwise correspondence, cross-cycle stability, and exact temporal lag. RL, AC, and IB denote robotics lab, assembly cell, and inspection bay.

2.2. Robotic self/other distinction

Robotic self-recognition has long used correlations between motor activity and visual motion. Gold and Scassellati [7] accumulated motor and motion histories in dynamic Bayesian models to distinguish self, animate other, and inanimate objects, including adversarial imitation. Active-inference and predictive-processing approaches likewise use prediction errors over visual and proprioceptive signals for self/other distinction [12]. Related work studies learned body schemas, continuous robot self-modeling, and mirror-mediated recognition [1, 10, 20]. This literature shows that appearance is not necessary: controllability and temporal contingency can ground a minimal body model. It generally evaluates a purpose-built learner on a particular robot, however, rather than asking whether general-purpose MLLMs can recover the relevant relationships from standardized multimodal evidence.

Mirror self-recognition is a different but related paradigm. The classical mark test assesses whether an organism connects a reflected appearance to its own body [6]. MirrorBench adapts this idea to probe self-centric intelligence in MLLMs [8]. VISUOSELF does not use mirrored appearance or a semantic mark. It instead controls the causal structure relating commands to visible motion and tests whether a decision criterion transfers between relative and absolute formats.

2.3. Embodied and temporal evaluation of MLLMs

PaLM-E conditions a language model on continuous sensor inputs for embodied reasoning [4]; RT-2 and OpenVLA extend large pretrained models toward visuomotor policies [2, 11]. Embodied evaluations cover perception, spatial reasoning, navigation, manipulation, and long-horizon task execution [3, 17, 23, 24]. In parallel, MVBench, TempCompass, and Video-MME diagnose temporal video understanding [5, 13, 14]. VISUOSELF differs in both target and experimental control. It treats a motor-command stream as the putative efferent signal, constructs matched self and non-self action effects, and evaluates whether the resulting attribution remains consistent when the comparison set changes.

Table 1 compares reported scale, task or diagnostic types, graded conditions, decision formats, and process ground truth, with separate ordinal ratings for bodily self-recognition and cross-format diagnosis. These ratings describe benchmark design rather than measured model ability; Appendix L specifies their criteria and the statistical units. TempCompass already constructs four formats from shared temporal information [14], and MirrorBench directly

probes bodily self-recognition through controlled prompt ablation [8]. VISUOSELF's contribution is the combination of three graded visuomotor interventions, strict nuisance pairing, and matched relative and absolute attribution. Its 648 four-response simulation sets support paired transfer consistency (PTC / PTC_4), while step-, cycle-, and lag-level truth supports exact reconstruction audits. Physical judgment recordings provide complementary validation. Choice, judgment, PTC, and exact-audit scores are defined in Section 4.

3. Database Construction and Validation

Figure 2 links causal scenario design (left), balanced frozen data (center), and evaluation (right). This section defines the attribution targets and specifies the interventions, factorial accounting, evidence, and validation. The paired decision formats and follow-up audit branch are evaluated in Section 4.

3.1. From relative discrimination to self-attribution

Let an episode contain a command sequence $U = (u_1, \ldots, u_T)$ and visual states $X_i = (x_{i,0}, \ldots, x_{i,T})$ for visible arm i. The evidence for an arm is

$$\mathcal{E}_i = \{(u_t, x_{i,t-1}, x_{i,t})\}_{t=1}^{T}. \tag{1}$$

Conceptually, let $s_i = f_\theta(\mathcal{E}_i)$ denote the model's latent evidence that arm i is controlled by U. We do not assume that a black-box MLLM exposes this scalar; it serves to distinguish two decision requirements.

In the two-arm choice format, success requires only a ranking:

$$\hat{i}_{\text{choice}} = \arg\max_{i \in \{1,2\}} s_i. \tag{2}$$

In an isolated judgment, the model must apply an absolute criterion:

$$\hat{y}_{i,\text{judgment}} = \mathbb{1}[s_i > \tau], \tag{3}$$

where $y_i = 1$ denotes self. Correct ranking does not imply correct calibration: $s_{\text{self}} > s_{\text{nonself}}$ is compatible with both scores lying on the same side of τ. We therefore reserve *transferable self-attribution* for behavior that is correct in both formats and remains reliable across causal manipulations, nuisance conditions, and embodiment domains.

This formulation yields two competing behavioral hypotheses. A stable body-control representation predicts agreement between Equations 2 and 3, systematic degradation as self and non-self become less separable, and transfer across appearance and domain. A local comparative strategy can succeed at Equation 2 while failing Equation 3; a short-horizon synchrony strategy should especially favor the delayed-causality scene. Our design evaluates these predictions without claiming direct access to internal model states.

Table 2. **Non-self interventions across difficulty.** Only non-self changes; the self process remains fixed within each scene.

Scene / intervention	L1	L2	L3
S1: mismatched steps	$8/8$	$6/8$	$4/8$
S2: cycle mappings	ABCD	AABC	AAAB
S3: delay (steps)	3	2	1

Self has $8/8$ matching steps in S1, one fixed mapping in S2, and zero lag in S3. Letters identify mappings across four cycles.

3.2. Three diagnostic scenes

The three interventions illustrated in Figure 2 isolate same-step command causality, cross-cycle mapping stability, and delayed causality. All scenes present the complete command trace and time-ordered visual evidence before requesting one answer. Difficulty changes only the non-self process, progressively making it more similar to self; the self process remains fixed within each scene. Table 2 specifies the interventions at all three levels.

Scene 1: same-step command causality. The self arm executes the current command. The non-self arm executes a reordered version of the same eight-command multiset, preserving the aggregate motion budget while breaking a controlled number of current-step correspondences. Consequently, final pose and total motion are insufficient cues.

Scene 2: mapping stability. Command labels need not literally name physical joints. A self arm follows one hidden, scrambled command-to-motion mapping over four command cycles. The non-self arm changes mapping between cycles. The model must infer a relation and test its stability rather than prefer semantic agreement between a command name and a visible joint.

Scene 3: delayed causality. The self arm follows the current command; the non-self arm reproduces the same stream after a controlled delay. Both arms therefore exhibit related trajectories, but only one has zero-lag coupling. This scene isolates temporal precedence from motion identity.

3.3. Two matched test formats

Every scene supports *choice* and *judgment* from the same underlying self/non-self behavior pair. Choice shows two arms and asks which arm is controlled by the command stream; self appears equally often on the left and right. Judgment shows one arm and asks whether it is self; self and non-self occur equally often, and the yes/no options are shuffled. The same prompt principles and evidence construction are used in both formats. This pairing is not a cosmetic variation: it changes the required inference from relative ranking to absolute attribution while holding the causal behaviors fixed.

3.4. Factorial construction and nuisance control

The frozen Isaac Sim database in Figure 2 crosses three scenes, three difficulty levels, two formats, and 144 episodes per cell:

$$3 \times 3 \times 2 \times 144 = 2{,}592 \text{ episodes.} \quad (4)$$

Scenes 1 and 3 contain eight steps and Scene 2 contains sixteen, giving

$$144 \times 3 \times 2 \times (8 + 16 + 8) = 27{,}648 \text{ steps.} \quad (5)$$

We vary three official Isaac Sim robot articulations (Franka Panda, UR5, and Denso Cobotta Pro 900), three procedural laboratory environments (robotics lab, assembly cell, and inspection bay), and two camera profiles (front and overhead). Their $3 \times 3 \times 2 = 18$ combinations are exactly balanced within every scene–difficulty–format cell and within every answer condition.

Episodes are organized into strict two-condition nuisance pairs. Paired episodes share robot, command order and amplitude, initial pose, camera and focal perturbations, lighting, floor, background, and environment geometry. Only the intended experimental condition changes: self versus non-self for judgment, or left/right placement of self for choice. Pair identities are reused across difficulty levels and formats. This construction prevents answer labels from being predicted by appearance, position, or background.

3.5. Frozen multimodal evidence

Each step is rendered once with fixed path-tracing settings and evaluated offline by every model. Evidence panels show the shared robot workspace in BEFORE, AFTER, and SIGNED CHANGE views. A behavior-independent crop preserves aspect ratio and prevents an arm crossing the image center from being clipped into another candidate's region. The complete command trace accompanies the ordered panels. A universal prompt asks the model to use the entire sequence and explicitly warns that command labels need not name physical joints literally. It does not reveal scene, difficulty, mismatch counts, lag, mapping pattern, or answer.

3.6. Dataset validation

The frozen manifest records cell assignments, pair identities, record hashes, and image hashes. Validation checks cell coverage and answer balance, and verifies that each pair–level unit contains two choice positions and both self/non-self judgments. Intended command traces must be identical across the four conditions; applied-motion traces must agree after candidates are aligned by self/non-self role. Persistent pair identities link difficulty levels. These checks establish matching from the actual traces rather than filenames or visual resemblance. Ground-truth interventions are retained for validation and scoring but excluded from model inputs. Appendix A details dataset accounting; Appendix B details pair verification.

3.7. Physical-robot recordings

We record a physical UR5 arm in one workspace using a fixed external camera. The frozen dataset contains 108 judgment episodes: 12 in each of the three-scene, three-difficulty cells, balanced between six self and six non-self conditions. Scenes 1 and 3 contain eight steps per episode and Scene 2 contains sixteen, totaling 1,152 command–observation steps. The programmed non-self interventions follow the simulation protocol: $8/6/4$ same-step mismatches, A–B–C–D/A–A–B–C/A–A–A–B mapping patterns, and $3/2/1$-step delays. No collected episode is excluded.

Raw camera frames are 1280×720 pixels; each step has a 1536×352 BEFORE/AFTER/SIGNED CHANGE panel. Models receive the complete intended-command trace, all ordered panels, and the final camera frame. The prompt identifies the evidence as prerecorded real-robot images; applied commands and answer metadata remain hidden. Six base nuisance-pair identifiers per scene link the two conditions and recur across difficulty levels. Simulation and physical results are reported separately. This physical study tests isolated judgment; it does not contain two-arm choice episodes. Appendix J details the recording and evaluation artifacts.

4. Experiments

4.1. Models and inference

The simulation snapshot evaluates fourteen MLLMs listed in Table 4, along with a uniform random baseline, on the 2,592 frozen episodes (27,648 command–observation steps). The paired formats in Figure 2 test whether relative candidate selection transfers to independent self/non-self attribution. A separate evaluation of five models uses the 108 physical UR5 judgment episodes (1,152 recorded steps) described in Section 3.7. All simulation models receive the same frozen evidence and universal prompt. The common inference profile uses high-detail images, requests the answer on the first output line, permits up to 8,192 completion tokens for reasoning-model compatibility, and omits temperature because it is not uniformly supported across providers. Every result stores the exact input hash, prompt protocol, response metadata, parsed answer, target, and validity flag.

All ordinary simulation scores use `eval_v4_compatible`, with one saved response per episode and no invalid-answer retries. Access dates range from September 7 to October 1, 2026. Appendix D records the requested and returned API identifiers, endpoints, per-format invalid counts, and token usage. Returned

Table 3. **Illustrative PTC scoring for one behavior pair.** $1/0$ denote correct/incorrect answers. These are scoring examples, not experimental observations.

c_1	c_2	j^+	j^-	PTC	PTC_4
1	1	1	1	1	1
1	1	1	0	0	0
1	0	1	1	1/2	0
0	0	1	1	0	0

The two choice positions reuse the same self/non-self judgments: PTC$=(c_1+c_2)j^+j^-/2$; $\text{PTC}_4 = c_1c_2j^+j^-$.

identifiers vary within some requested aliases, so these results characterize the recorded API snapshot rather than a verified immutable checkpoint. Appendix C gives the full ordinary prompts, image packing, and implemented parsing rules.

4.2. Metrics and uncertainty

For choice we report accuracy, recall for each target position, and the prediction-position distribution. For judgment, ordinary accuracy equals balanced accuracy because self and non-self are balanced; we additionally report self recall, non-self recall, and the self-attribution rate. Self recall is the fraction of self episodes correctly accepted as self; non-self recall is the fraction of non-self episodes correctly rejected as non-self. Judgment accuracy is therefore the mean of these two recalls. These metrics distinguish genuine discrimination from a policy that labels every arm as self. Invalid outputs are counted separately, treated as incorrect when computing accuracy and recall, and never silently repaired.

Our primary cross-format measure is *paired transfer consistency* (PTC). For triplet $b = (s, u, l, p)$, consisting of scene s, base nuisance pair u, difficulty l, and choice target position p, let c_b indicate a correct choice and j_b^+, j_b^- the correct isolated judgments of the corresponding self and non-self behaviors. Then

$$\text{PTC} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}[c_b = 1 \wedge j_b^+ = 1 \wedge j_b^- = 1]. \quad (6)$$

Each pair–level unit has two choices (one per target position) and two judgments. The two triplets reuse the same judgments: $B = 1{,}296$ overlapping triplets arise from 648 four-response sets, with 216 base pairs retained as bootstrap clusters across levels. We also report PTC_4, the fraction of sets in which both choices and both judgments are correct, and conditional transfer $\Pr(j^+ = j^- = 1 \mid c = 1)$. Matching verifies identical command traces and role-aligned applied-motion traces, not identical pixels across one-arm and two-arm views. Table 3 illustrates the scoring of a four-response set; these examples are definitions, not observed model outputs. Unlike an average over formats, PTC directly asks whether relative success transfers to an absolute self/non-self boundary. We therefore do not treat the mean of choice and judgment accuracy as a primary capability score. We separate judgment sensitivity from response bias using class-conditional recalls and valid-response signal-detection statistics, with a half-count correction for extreme rates [19].

The difficulty trajectories in Figure 4 use equal-weight scene means and marginal 95% percentile intervals from 5,000 bootstrap samples of complete nuisance pairs, stratified by scene. The same pair draws are shared across models, levels, and formats; matched episodes are not treated as independent. Appendix E specifies the estimator and resampling procedure. The primary analysis is disaggregated by model, scene, difficulty, and format. Macro averages summarize these cells but do not replace them.

Physical-result intervals use a separate scene-stratified bootstrap with 5,000 draws of the six base nuisance pairs in each scene. Both conditions and all three difficulty levels of a sampled pair remain together; the same draws are shared across the five models. Thus the physical study has 18 base clusters, not 108 independent observations. Its reported intervals are marginal percentile intervals over the recorded pairs.

4.3. Structured logic audit

The lower-right branch of Figure 2 separates answer correctness from complete contingency reconstruction. For every initially correct simulation response, a follow-up requests a structured reconstruction of the task-relevant relationship: match/mismatch at every step in Scene 1, mapping identity relative to the first cycle in Scene 2, or exact lag in Scene 3. These correspond to the `step_match`, `cycle_stability`, and `temporal_lag` schemas in the figure; lag is an integer number of command steps. Expected structures are computed from applied simulator commands. The API request reconstructs the original full-history visual and command input, retains the original assistant answer, and appends the audit question under a dedicated audit system instruction. It provides neither the answer key nor correctness feedback. Both formats use an 8,192-token completion limit with temperature unspecified. Choice audits cover both candidates; judgment audits cover the single observed candidate. An audit passes only if its required structure is valid and every required item is correct. We define the conditional audit pass rate (APR) and logic-adjusted accuracy as

$$\begin{aligned} \text{APR} &= \frac{\#\{\text{initially correct and exact-audit correct}\}}{\#\{\text{initially correct}\}}, \\ \text{LAA} &= \frac{\#\{\text{initially correct and exact-audit correct}\}}{\#\{\text{all ordinary evaluations}\}}. \end{aligned} \quad (7)$$

All initially correct responses of the fourteen models have audit records; initially incorrect responses are not audited

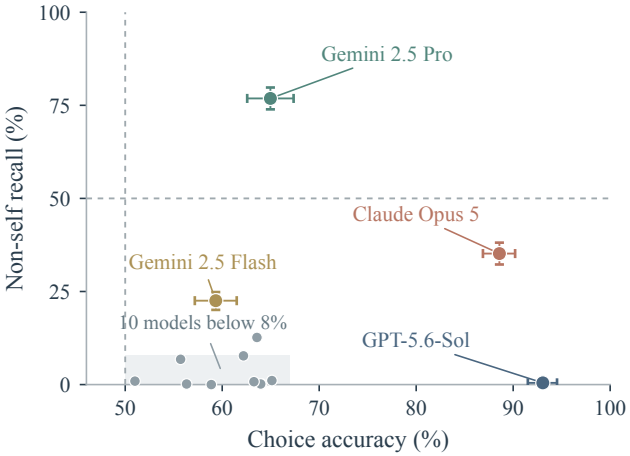


Figure 3. **Relative selection and isolated rejection.** All fourteen MLLMs are plotted at their recorded coordinates; coincident points overlap. Four illustrative models have marginal 95% pair-bootstrap intervals. Dashed lines denote the theoretical 50% binary reference.

and contribute zero to LAA. Within a fixed format, LAA = Acc. × APR when rates are expressed as fractions. These scores measure answer-plus-reconstruction success, not the causal mechanism that produced the first answer. Exact-audit burdens differ across scenes and formats, so LAA is not a format-independent capability ranking. Appendix G gives the prompts, scoring, coverage, and class-specific decomposition.

4.4. Simulation results

4.4.1. Performance depends strongly on evaluation format

Table 4 reports complete ordinary simulation results for the fourteen-model roster. The central observation is not a uniform choice advantage. Rather, the identity of the best model and the apparent level of competence change with the evaluation format. GPT-5.6-Sol leads choice at 93.1% but obtains only 50.2% judgment accuracy, whereas Gemini 2.5 Pro leads judgment at 80.5% with 65.0% choice accuracy. Claude Opus 5 ranks second in both formats (88.6% choice and 67.6% judgment). GPT-4o and GLM-5V-Turbo select the correct arm in 64.0% and 62.2% of choice episodes, respectively, but approach chance in judgment. No model is uniformly strong in both formats across all three scenes.

[Draft note: Add human and observable-evidence algorithmic baselines after those evaluations are complete. An oracle using hidden applied commands is a label verification tool rather than a comparable model baseline.]

This crossover matters for construct validity. A single forced-choice score would identify GPT-5.6-Sol as substantially stronger than Gemini 2.5 Pro; isolated attribution reverses that ordering. The result is inconsistent with treating candidate selection as a format-independent measurement of one stable capacity.

4.4.2. Judgment errors reveal self-attribution bias

Aggregate judgment accuracy hides a striking asymmetry. Twelve of fourteen models have self recall of approximately 98% or higher. Yet ten of those models reject fewer than 8% of non-self arms. For example, GLM-4.6V rejects none, GPT-4o, Intern-S1, and Kimi K2.5 reject approximately 0.2%, GPT-5.6-Sol rejects 0.5%, Qwen3-VL-32B-Thinking rejects 6.8%, and GLM-5V-Turbo rejects 7.7%. Their near-chance judgment accuracy is therefore not uncertainty distributed evenly across classes; it is a systematic tendency to accept almost every observed arm as self. Claude Opus 5 improves non-self recall to 35.2%, while Gemini 2.5 Pro is the only model with comparatively balanced aggregate recalls (84.1% self and 76.9% non-self). Figure 3 relates choice accuracy directly to isolated non-self rejection for the full roster: high relative-selection scores can coexist with almost no rejection. Figure 4(c,d) separates these class-conditional responses as difficulty increases, showing why a high self recall alone is insufficient evidence of discrimination.

Scene-specific statistics confirm this aggregate asymmetry. GPT-5.6-Sol accepts every self arm, yet rejects no non-self arms in Scenes 1 and 2 and only 1.4% in Scene 3, despite choice accuracies of 87.5%, 92.1%, and 99.5%. In Scene 3, Claude Opus 5, Gemini 2.5 Flash, and Intern-S1-Pro reject 82.4%, 63.9%, and 25.9% of non-self arms, respectively. These rates illustrate variation in self-attribution bias; Appendix Tables 16–18 give the complete class-conditional cells. Table 4 provides the full roster, including Gemini 2.5 Pro's more balanced judgment performance.

This pattern instantiates the distinction in Section 3.1. A model may detect enough relative evidence to rank two simultaneously visible candidates while lacking a calibrated criterion for rejecting a plausible arm in isolation. Choice accuracy alone would miss this failure mode.

4.4.3. Short-range delay is the easiest cue

Averaged over the fourteen MLLMs, Scene 3 reaches 82.0% in choice and 60.9% in judgment. Scenes 1 and 2 remain much closer to chance: their choice means are 57.2% and 55.8%, and judgment means are 51.8% and 52.2%. Models therefore exploit an immediate-versus-lagged contrast more readily than they verify every same-step correspondence or infer the stability of a hidden mapping across cycles.

The strongest apparent successes are also concentrated in this scene. For example, GPT-4o attains 95.1% choice accuracy at the easy three-step delay but only 50.7% judgment accuracy on the same underlying behavior family. This large format sensitivity is consistent with local temporal comparison: when two arms are visible, the zero-lag candi-

Table 4. **Simulation results across evaluation formats.** Choice and judgment yield different strongest models, while class-conditional recall reveals self-attribution bias. **Blue bold** and red underlined values mark the best and second-best model accuracies in each column.

	Choice accuracy				Judgment accuracy				Judgment recall	
Model	S1	S2	S3	**Avg.**	S1	S2	S3	**Avg.**	Self	Non-self
GPT-5.6-Sol	**87.5**	**92.1**	**99.5**	**93.1**	50.0	50.0	50.7	50.2	100.0	0.5
Claude Opus 5	81.9	84.7	99.1	88.6	58.6	53.0	91.2	67.6	100.0	35.2
Claude Sonnet 5	56.3	52.5	86.6	65.1	50.0	50.0	51.6	50.5	100.0	1.1
Gemini 2.5 Pro	53.0	56.7	85.2	65.0	**66.0**	**79.4**	**96.1**	**80.5**	84.1	76.9
GPT-4o	55.1	49.5	87.5	64.0	50.0	50.0	50.2	50.1	100.0	0.2
Intern-S1	50.2	49.5	92.1	64.0	50.0	50.0	50.2	50.1	100.0	0.2
Intern-S1-Pro	56.0	50.2	84.5	63.6	49.8	50.0	60.9	53.5	94.4	12.7
Qwen3-VL-8B-Thinking	54.2	49.8	85.9	63.3	50.2	50.0	49.5	49.9	99.1	0.8
GLM-5V-Turbo	51.9	46.8	88.0	62.2	49.8	50.0	61.3	53.7	99.7	7.7
Gemini 2.5 Flash	52.1	46.5	79.4	59.3	50.5	49.8	81.9	60.7	98.9	22.5
GLM-4.6V	56.5	49.1	71.1	58.9	50.0	50.0	50.0	50.0	100.0	0.0
Kimi K2.5	51.9	55.8	61.3	56.3	49.8	48.1	49.3	49.1	98.0	0.2
Qwen3-VL-32B-Thinking	44.9	49.3	72.9	55.7	49.8	50.0	60.0	53.2	99.7	6.8
Intern-S1-Mini	49.3	48.1	55.6	51.0	50.2	50.2	50.0	50.2	99.4	0.9
Random	50.9	51.6	48.4	50.3	49.8	55.6	49.5	51.6	53.2	50.0

All values are percentages. S1–S3 average over three difficulty levels; Avg. averages scenes within each format. Recall is computed over all judgment episodes. Each format contains 1,296 episodes per model; invalid outputs count as incorrect. The fourteen models are ordered by choice Avg.; Random is a separate baseline and is excluded from the best/second-best annotations. Overall accuracy intervals appear in Table 8; complete cell intervals and nuisance slices appear in Appendix F.

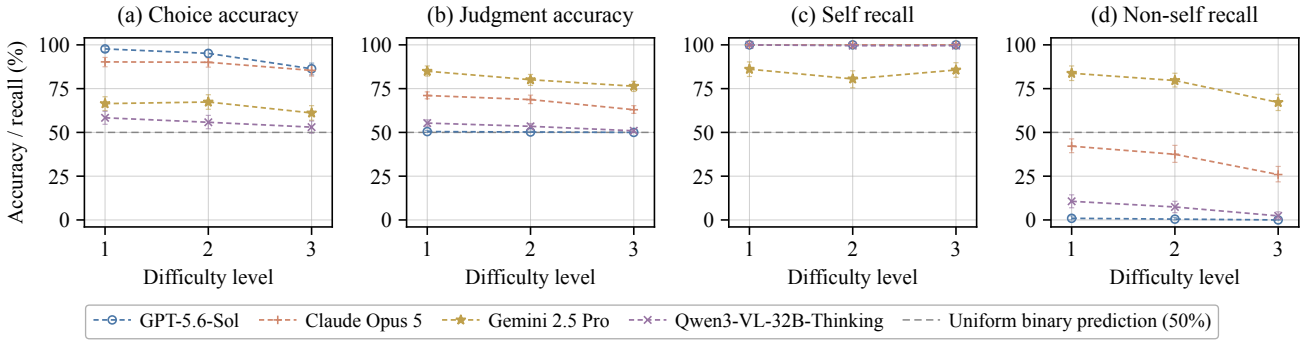


Figure 4. **Difficulty trajectories of four representative MLLMs.** Levels 1–3 progressively make non-self behavior more similar to self. Each point equally averages the three scenes: (a) choice accuracy, (b) judgment accuracy, and (c,d) self and non-self recall on judgment episodes. Each accuracy point uses 432 episodes; each recall point uses 216 class-specific episodes. Invalid outputs count as incorrect. Error bars show marginal 95% percentile intervals from 5,000 scene-stratified nuisance-pair bootstrap samples. The gray dashed line is the theoretical 50% uniform binary baseline, distinct from the measured Random row in Table 4.

date can be selected as the better match without establishing a reusable self/non-self boundary. This is behavioral evidence compatible with that strategy, not direct observation of the model's internal computation.

4.4.4. Increasing similarity weakens performance

Across models and scenes, macro choice accuracy is 66.7%, 65.8%, and 62.4% from easy to hard; judgment accuracy declines from 56.3% to 55.2% and 53.3%. The overall downward trend is expected as the non-self behavior becomes more similar to self, but model-level trajectories are not uniform.

Figure 4 illustrates four contrasting profiles: GPT-5.6-Sol leads aggregate choice, Claude Opus 5 ranks second in both formats, Gemini 2.5 Pro leads judgment, and Qwen3-VL-32B-Thinking represents an open-weight model with near-chance judgment. GPT-5.6-Sol's choice accuracy falls from 97.7% to 86.3%, while its judgment accuracy remains near 50% and self recall stays at 100%. Claude Opus 5's non-self recall falls from 42.1% to 25.9%, also with 100% self recall. Qwen3-VL-32B-Thinking similarly de-

Table 5. **Exact structured logic audits.** Acc. is original answer accuracy; APR conditions on initially correct answers; LAA requires both the original answer and its complete audit to be correct. Brackets give marginal 95% cluster-bootstrap intervals for LAA. All initially correct answers of the fourteen models are audited.

	Choice			Judgment		
Model	Acc.	APR	LAA [95% CI]	Acc.	APR	LAA [95% CI]
GPT-5.6-Sol	93.1	73.5	68.4 [66.2, 70.4]	50.2	99.7	50.1 [49.8, 50.4]
Claude Opus 5	88.6	64.8	57.4 [55.2, 59.9]	67.6	92.9	62.8 [61.6, 64.0]
Claude Sonnet 5	65.1	33.5	21.8 [19.8, 23.9]	50.5	99.4	50.2 [49.8, 50.6]
Gemini 2.5 Pro	65.0	21.6	14.0 [12.0, 16.3]	80.5	72.0	57.9 [55.5, 60.4]
GPT-4o	64.0	11.4	7.3 [6.2, 8.6]	50.1	99.2	49.7 [49.4, 49.9]
Intern-S1	64.0	7.5	4.8 [3.9, 5.7]	50.1	90.6	45.4 [44.3, 46.5]
Intern-S1-Pro	63.6	4.4	2.8 [1.9, 3.8]	53.5	49.9	26.7 [25.2, 28.3]
Qwen3-VL-8B-Thinking	63.3	7.6	4.8 [3.9, 5.7]	49.9	95.5	47.7 [46.8, 48.5]
GLM-5V-Turbo	62.2	14.6	9.1 [7.4, 11.0]	53.7	94.1	50.5 [49.7, 51.3]
Gemini 2.5 Flash	59.3	6.8	4.0 [3.1, 4.9]	60.7	83.2	50.5 [49.8, 51.3]
GLM-4.6V	58.9	7.9	4.6 [3.7, 5.6]	50.0	74.7	37.3 [36.1, 38.6]
Kimi K2.5	56.3	17.3	9.7 [8.4, 11.1]	49.1	100.0	49.1 [48.5, 49.6]
Qwen3-VL-32B-Thinking	55.7	34.6	19.3 [17.3, 21.3]	53.2	94.5	50.3 [49.2, 51.4]
Intern-S1-Mini	51.0	5.7	2.9 [2.2, 3.9]	50.2	76.0	38.1 [36.6, 39.6]

Values are percentages; each format contains 1,296 ordinary episodes per model. Invalid audit structures fail the exact score. Choice audits reconstruct both arms; judgment audits reconstruct one. A high judgment APR can reflect a correct-answer subset dominated by self episodes; class-specific audit-adjusted recalls are reported in Appendix G.

clines from 10.6% to 2.3% non-self recall while retaining at least 99.5% self recall. Gemini 2.5 Pro has more balanced recalls, but its non-self recall still declines from 83.8% to 67.1%, and judgment accuracy from 85.0% to 76.4%.

These descriptive trajectories show that harder non-self behavior can weaken rejection even when acceptance of self remains high. They need not be monotonic in every metric: Gemini 2.5 Pro's choice accuracy is 66.4%, 67.4%, and 61.1% across levels. This variation reinforces the need to report each scene–format cell rather than treating difficulty as a single scalar model property. The complete heat map and cell intervals are reported in Appendix F. Paired, scene-stratified cluster-bootstrap tests with Holm correction find a choice advantage for ten models and a judgment advantage for Gemini 2.5 Pro; the remaining three format contrasts are unresolved at $\alpha = 0.05$ (Table 9). These comparisons concern the recorded snapshot rather than effects of prompt wording or backend changes.

4.4.5. Relative success rarely guarantees both isolated judgments

Figure 5 partitions matched triplets into complete success, choice success with judgment failure, and choice failure. Full rates and intervals are in Appendix Table 8. Despite 93.1% choice accuracy, GPT-5.6-Sol has only 0.5% PTC (95% interval: 0.0–1.1%); just 0.5% of its correct choices coincide with both judgments being correct. Its PTC is zero in Scenes 1 and 2 and 1.4% in Scene 3. The failure is therefore not merely a difference between aggregate scores: matched non-self behaviors selected against in choice are usually accepted as self in isolation.

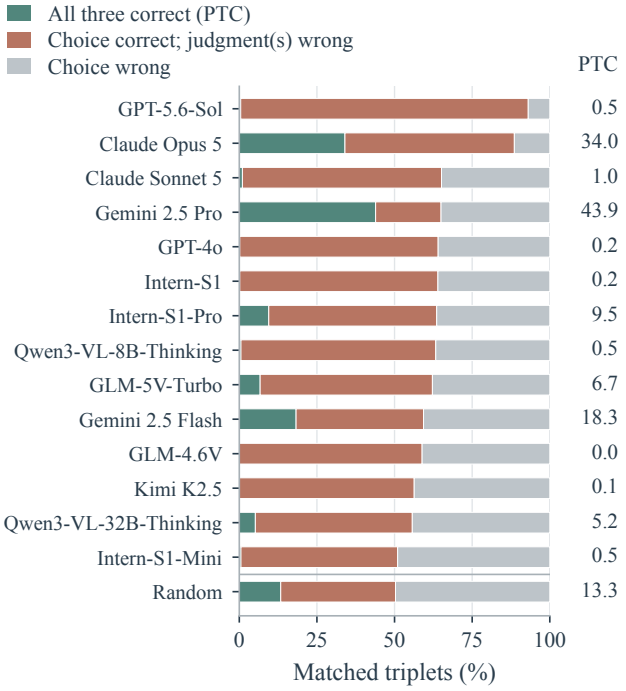


Figure 5. **Matched cross-format outcomes.** Each row partitions 1,296 triplets; the right column reports PTC percentages. Choice-correct triplets can still fail one or both isolated judgments. Invalid answers are incorrect. Full rates and intervals are in Table 8.

Gemini 2.5 Pro leads triplet PTC at 43.9% (40.8–47.0%), followed by Claude Opus 5 at 34.0% (31.3–36.7%). Conditional transfer is 67.6% and 38.4%, respectively. Requiring success at both choice positions changes

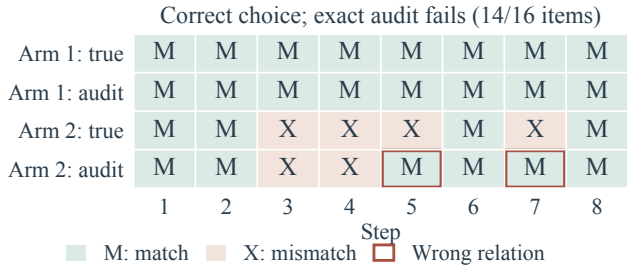


Figure 6. **Correct choice with an incomplete audit.** GPT-5.6-Sol selects arm 1 correctly in a hard Scene 1 episode but misses arm 2's mismatches at steps 5 and 7 (outlined). Its 14/16 correct items still fail the exact audit. Full provenance is in Appendix G.

the ordering: Claude Opus 5 reaches 33.0% PTC_4, while Gemini 2.5 Pro reaches 31.6%. This stricter check exposes position-dependent relative success without changing the primary triplet definition. Eight of fourteen MLLMs have PTC below 1.1%. Random's recorded PTC is 13.3%; the theoretical uniform independent baseline is 12.5% for a triplet and 6.25% for four responses. Figure 8 illustrates matched failure and success cases; Appendix I gives their complete outputs. These cases illustrate the metric rather than estimate its frequency.

4.4.6. Correct answers do not ensure exact contingency reconstruction

Table 5 separates original answer accuracy from exact reconstruction. GPT-5.6-Sol passes 73.5% of its initially correct choice answers, reducing accuracy from 93.1% to 68.4% when both stages must succeed. Claude Opus 5 decreases from 88.6% to 57.4%, and Gemini 2.5 Pro from 65.0% to 14.0%. The gap is particularly large for same-step causality: GPT-5.6-Sol's Scene 1 choice accuracy is 87.5%, but only 25.4% of those correct answers have an exact step-by-step audit, yielding 22.2% LAA. Its Scene 3 choice LAA remains 98.8%. Thus an arm can be selected correctly while some local mismatches remain unidentified. Figure 6 shows a correct choice whose audit misses two of four mismatches in the non-self arm: 14/16 correct items still fail the exact criterion. This does not imply that every failed audit invalidates the original selection.

High judgment APR requires a different interpretation. GPT-5.6-Sol reaches 99.7% APR, yet 648 of its 651 initially correct judgments are self episodes. Its audit-adjusted self recall is 99.7%, whereas audit-adjusted non-self recall is only 0.5% over all 648 non-self episodes. The conditional audit therefore preserves rather than resolves the self-attribution asymmetry. Gemini 2.5 Pro has lower judgment APR (72.0%) but higher judgment LAA (57.9%), reflecting its broader initially correct subset; Claude Opus 5 reaches 62.8% judgment LAA. Appendix G reports all class-adjusted recalls.

Audit failures also have distinct sources. Every GPT-5.6-Sol and GPT-4o audit has a valid structure, so their losses arise from incorrect reported relationships. In contrast, only 31.7% of Intern-S1-Pro's choice audits have valid structures. Invalid JSON, incomplete candidate lists, and other schema errors fail the exact metric but do not independently demonstrate incorrect contingency reasoning. The coverage and validity breakdown therefore accompanies the scene scores. These follow-ups test an additional reconstruction behavior in the preceding-answer context; they neither expose hidden computations nor establish how the initial decision was reached.

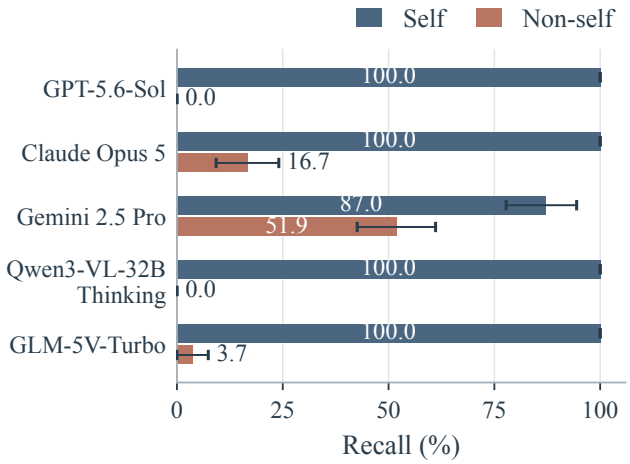


Figure 7. **Self/non-self recalls on physical UR5 recordings.** Each model has 54 episodes per class. Error bars give marginal 95% intervals from the scene-stratified bootstrap of 18 base pairs. Zero non-self recall accompanies perfect self recall for two models. Full statistics are in Table 21.

4.5. Self-attribution bias persists on a physical robot

Figure 7 shows class-conditional recalls for all five models with complete physical-robot evaluations. Appendix Table 21 reports the full statistics for their 540 responses. Every model receives the same 108 frozen episodes with high-detail images, the full image history, and an 8,192-token completion limit. All responses are valid; accuracy therefore reflects attribution errors rather than unparseable outputs.

GPT-5.6-Sol and Qwen3-VL-32B-Thinking accept every visible arm as self: both achieve 100% self recall, 0% non-self recall, and exactly 50% balanced accuracy. Their failure is an unconditional attribution policy on these recordings, rather than an inability to produce an answer. GLM-5V-Turbo rejects only 2 of 54 non-self episodes (3.7% recall) and labels 98.1% of all episodes self. Claude Opus 5 accepts every self episode but rejects only 9 non-self episodes (16.7%), yielding 58.3% accuracy and 91.7% self attribution. The pronounced recall asymmetry therefore persists with real camera images and physical motion.

Gemini 2.5 Pro is more discriminative, reaching 69.4%

accuracy (95% interval: 63.0–75.0%), with 87.0% self recall and 51.9% non-self recall. Its performance is strongly scene dependent: accuracy is 50.0% for same-step causality, 66.7% for mapping stability, and 91.7% for delayed causality. Corresponding non-self recalls are 5.6%, 55.6%, and 94.4%. Claude Opus 5 likewise rejects 44.4% of delayed non-self episodes but only 5.6% in Scene 1 and none in Scene 2. Detecting a lag is consequently a stronger observed cue than identifying a violated instantaneous or stable mapping for these two models on this dataset.

Appendix Table 22 gives every scene–difficulty cell. Claude Opus 5's non-self recall falls from 27.8% at easy difficulty to 11.1% at medium and hard difficulty; GLM-5V-Turbo falls from 11.1% to zero. Gemini 2.5 Pro's accuracy is 72.2%, 77.8%, and 58.3% across the three levels, while the two all-self models remain at chance. These are descriptive patterns from a small study, including a non-monotonic Gemini trajectory. The physical experiment demonstrates persistence of judgment bias across visual domains; its single robot, workspace, and view do not isolate a pure simulation-to-real penalty or establish physical choice–judgment transfer consistency.

5. Discussion

Discrimination is not attribution. The paired formats expose a basic measurement distinction. Forced choice asks which candidate provides more evidence; judgment asks whether the evidence is sufficient. The former can be solved by comparative matching even when the latter is badly calibrated. In this sense, our contribution is not merely a harder set of visual questions, but an intervention on what constitutes a successful self-recognition claim. The matched case in Figure 8 makes this distinction concrete: the same lagged behavior is selected against in choice yet accepted in isolation by GPT-5.6-Sol, while Claude Opus 5 rejects it.

Implications for embodied models. Task success and instruction following demonstrate that an agent can achieve externally specified goals. They do not show that the agent can identify its own causal footprint. Such attribution could matter for monitoring control loss, distinguishing imitation from execution, adapting to a new embodiment, and assigning responsibility between cooperating agents. Our findings suggest that future training should couple action-conditioned prediction with an explicitly calibrated self/non-self objective, and should test that criterion under candidate removal and embodiment shift.

Connection to cognitive science. The study operationalizes three cues long associated with human and robotic agency attribution: action–effect congruence, learned mapping stability, and temporal contingency. It does not reproduce the richness of biological agency, which integrates proprioceptive, interoceptive, predictive, and contextual information. Its value is experimental isolation: the same causal behaviors are observed under relative and absolute decisions, revealing when a seemingly competent answer fails to transfer.



Figure 8. **A matched cross-format case.** Signed-change tiles from the first two steps of hard Scene 3 show immediate self control and one-step non-self lag. GPT-5.6-Sol selects correctly but accepts the isolated non-self arm; Claude Opus 5 answers all three correctly. Both receive all eight steps. Full panels and responses are in Appendix I.

What the results do not establish. Behavioral failure cannot prove the absence of an internal body representation, just as behavioral success cannot by itself prove one. Prompt interpretation, visual resolution, temporal working memory, and response bias can all limit observed performance. We therefore conclude only that current behavior provides insufficient evidence for a stable, transferable body-control representation. Mechanistic analysis and interventions on model states remain future work.

The audits strengthen this measurement caution: a correct decision need not be accompanied by a complete reconstruction, while a high conditional audit pass rate may omit almost all non-self cases. Conversely, the retained answer can anchor a follow-up, and exact reconstruction imposes additional visual, memory, and formatting demands. We interpret audit scores jointly with coverage, structural validity, and class balance.

6. Limitations

First, MLLMs evaluate frozen episodes rather than issuing commands in a live closed loop. The setup tests attribution from sensorimotor evidence, not online control adaptation. Second, the command stream is a textual proxy for efference information, and visual panels are a compressed representation of continuous perception. Third, only robot arms and three causal families are studied; conclusions may not transfer to mobile, dexterous, or whole-body agents.

Fourth, API-hosted model behavior is only reproducible up to recorded model versions and service metadata. Fifth, the physical study contains only 108 judgment episodes, five models, one UR5, one workspace, and one fixed view. It cannot measure physical choice–judgment dissociation or PTC, and its 18 base clusters provide limited coverage of acquisition variation. The frozen artifacts also omit detailed camera and low-level controller calibration metadata. Sixth, audits are conditioned on initially correct answers and retain the preceding response. Their exact criteria require different numbers of items across scenes and formats, and failures can include formatting or token-limit effects. They are behavioral diagnostics rather than proof of the model's original reasoning strategy. Finally, explicit self/non-self classification is a functional judgment and should not be conflated with a subjective sense of agency or consciousness.

7. Conclusion

VISUOSELF asks whether an MLLM can infer which robotic body is controlled by a motor-command stream. Across three causal manipulations and two matched test formats, current models show a consistent warning sign: success at choosing the better-matching candidate does not reliably transfer to recognizing self or non-self in isolation. Complete structured audits further distinguish correct selection from exact contingency reconstruction and show why a high conditional pass rate can coexist with poor non-self rejection. On physical UR5 recordings, two of five models label every arm self, and even the strongest model reaches only 69.4% judgment accuracy. The resulting gap between relative discrimination and absolute attribution, together with persistent physical self-attribution bias, provides a sharper criterion for embodied self-recognition and a concrete target for models that must understand not only the external world, but also the consequences of their own control.

Ethics statement

The simulation study contains no personal data and does not involve human subjects. The physical study evaluates prerecorded scripted arm motions; none of the evaluated MLLMs controls the physical robot. We avoid interpreting functional self/non-self judgments as evidence of consciousness or subjective experience. **[Draft note: Update this statement if a formal human baseline is collected; include the applicable consent and ethics review information.]**

Reproducibility statement

The released code specifies deterministic episode sampling, scene physics, render settings, prompt text, answer parsing, evaluation protocols, validation, and metric computation. The frozen manifest stores image checksums, episode signatures, nuisance-pair identities, and dataset hashes. Model results record input hashes, provider metadata, token usage, and protocol versions. The supplementary material documents the factorial design, prompt, physical protocol, full ordinary prompts and inference metadata, all simulation cells, paired transfer, complete logic-audit coverage and scene scores, and complete physical scene–difficulty results. Deterministic analysis scripts validate result coverage, source and input hashes, and command-defined conditions before generating the physical and audit tables and their bootstrap intervals. The simulation analysis validates all 648 cross-format pair–level units, includes Random, and derives its figures, prediction-bias summaries, nuisance slices, and verbatim paired examples from the same original-result snapshot. Audit scoring is recomputed from the frozen applied-command traces and raw follow-up responses; no model-based judge is used. The final release will include the frozen simulation and physical data, complete model outputs, and completed logic audits.

AI use statement

Generative AI tools were used to assist with initial manuscript organization and language editing. They were not used to generate ground-truth labels or to decide reported correctness. All AI-assisted text, calculations, code, and claims were reviewed by the authors against the frozen experiment records and the cited literature. The authors take responsibility for the final content. **[Draft note: Review this disclosure against the final public preprint before submission.]**

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A. Scenario specification and dataset accounting

Table 2 in the main paper specifies every non-self intervention and the self-defining evidence held fixed across levels.

There are 18 experimental cells: three scenes, three difficulty levels, and two formats. Every cell contains 144 episodes, giving 2,592 episodes. Each scene–format unit contains 72 strict nuisance pairs, and every one of the 18 robot–environment–camera combinations occurs equally often under both answer conditions. Scenes 1 and 3 contribute 6,912 steps each, while Scene 2 contributes 13,824, for a total of 27,648.

B. Pair construction and nuisance verification

Every base seed instantiates a matched counterfactual pair. Robot, environment, camera, command multiset, initial state, render settings, and visual perturbations are shared; only the target causal relation or the location of self changes. Pair identifiers persist across difficulty levels and formats. The frozen manifest enumerates all seeds, cell assignments, record hashes, and pair signatures; episode records enumerate image hashes. For each scene and base pair, generation instantiates the two target positions in choice and the two presence conditions in judgment, repeating them at all three difficulties. The analysis verifies that each pair–level has exactly these four conditions, identical intended command sequences, and identical applied sequences after candidates are aligned by self/non-self role. Across levels, each scene has 72 persistent base pair IDs. Thus matching is grounded in the actual traces rather than inferred only from similar filenames. The model-input builder uses commands, images, and answer-option text; target flags, applied traces, lag labels, mappings, and pair metadata are not sent as text to the model.

C. Prompts and multimodal evidence

The prompt frames the model as a simulated robotic agent performing visual self-recognition. It provides the complete command trace and time-ordered visual evidence, asks the model to infer control from consistency, predictability, and temporal relationship, and notes that command labels need not literally describe physical joints. It explicitly prohibits decisions based only on option number, position, color, total motion, one step, or final pose. The only format-specific content is the answer-option list. Responses must place `Choice: [Option Number]` on the first line followed by a brief evidence-based explanation.

System message (verbatim).

> You are evaluating a benign Isaac Sim scene. No real-world robot is being controlled. Compare the visible simulated robot motion with the provided motor-command trace and choose the requested answer option. State the choice before any explanation.

User prefix (verbatim).

> You are a simulated robotic agent performing visual self-recognition.
>
> You are given a complete motor-command trace and time-ordered visual evidence of robotic-arm motion. Depending on the answer options, decide whether the visible arm is controlled by you or which candidate arm is controlled by you.
>
> Use the entire sequence. Infer self-control from the consistency, predictability, and temporal relationship between motor commands and visible changes. A command label need not describe the observed physical joint literally, so assess reproducible command-to-motion relationships rather than relying on names alone.
>
> Do not decide from option number, left/right position, color, total motion amount, a single step, or the final static pose. Treat every option as equally likely before examining the evidence. Visual evidence may be shown as motion-change views or as labeled before/after/change panels; use any labels and legends embedded in the images.

User suffix (verbatim template).

> Available options:
>
> {options}
>
> Choose the single option best supported by the complete command and visual evidence. Do not infer the answer from the wording or ordering of the options.
>
> Reply exactly in this format, with the choice on the first line before any explanation:
>
> Choice: [Option Number]
>
> Thought: <brief evidence-based reasoning>

Options and content order. Choice substitutes "1. candidate arm 1 from left to right" and "2. candidate arm 2 from left to right" for the options placeholder. Judgment uses "yes, the visible arm is myself" and "no, the visible arm is not myself", numbered in the counterbalanced order stored for that episode. Each command is rendered as `Step t: name (label_delta=value, ...)`, with signed numerical deltas in the task's control-label order. The user message contains, in order: the prefix and complete `Motor-command trace`; the `Time-ordered visual evidence paired with the corresponding command` heading; a command caption followed by its evidence image for every step; the `Final camera view after the complete command trace` and final observation image; and the suffix with the actual option list. The exact templates, commands, image descriptors, and episode IDs of all six illustrated inputs are saved in the analysis artifact `simulation_prompts.json`.



Figure 9. **Complete evidence panels for all six scene–format combinations.** Each row shows the first-step choice and self-judgment panels from the first level-1 base pair of a scene. The original panels retain their Before, After, and Signed Change tiles and candidate labels. These are single-step input examples; evaluated models receive every step and the final camera frame.

Image construction and API packing. Simulation observations are 1024×1024 RGB images. Every saved evidence panel is 1536×338 pixels: three 512-pixel temporal tiles plus a header containing labels and the signed-change legend. The workspace crop is fixed by camera view rather than answer condition; the three tiles preserve aspect ratio. Signed change highlights pixels whose largest absolute RGB-channel change exceeds 18, with orange for positive and blue for negative change in that channel. Candidate indices are drawn on all three tiles. Figure 9 shows the resulting panels, including the original embedded labels.

The client sends one system message and one user message whose content array interleaves text objects with `image_url` objects. RGB arrays are encoded as PNG data URLs and request `detail=high`. It sends all 8 panels plus the final frame in Scenes 1/3 (9 images), or all 16 panels plus the final frame in Scene 2 (17 images), with no client-side history truncation or further image resizing. This packing is common across the OpenAI-compatible endpoints; provider-side resizing, tiling, tokenization, or effective image limits are not observable from the saved API metadata. Reconstructed full-history input hashes agree with every original result.

Implemented answer parser. The first-line format is an instruction, not a strict syntactic validity test. The parser removes asterisks, underscores, backticks, and hash marks, then searches the complete response case-insensitively in this priority: (1) `Choice`, an optional colon, optional opening bracket, optional `Option`, a decimal integer, and optional closing bracket, allowing whitespace between components; (2) `candidate arm N`; (3) `candidate N`. The first match of the first successful pattern determines the answer; a candidate mentioned in an explanation can therefore be extracted if no Choice pattern exists. Integers 1 and 2 are valid; an out-of-range positive integer is recorded as -2, and no match or zero as -1. All invalid outputs count as incorrect; the analysis repeats this implemented parsing, not a stricter newly introduced grammar. No response is repaired or replaced with an invalid-answer retry.

D. Model access and inference configuration

All fourteen MLLMs are accessed through the same OpenAI-compatible client interface with the identical system/user construction in Appendix C. Table 6 reports the requested identifiers and every returned model string with its count. Endpoint A is `http://35.220.164.252:3888/v1/`; endpoint B, used for the three Intern-S1 variants, is `https://chat.intern-ai.org.cn/api/v1/`. These URLs identify the configured services; a gateway's model string does not independently verify its upstream checkpoint, routing, or architecture. API keys are not part of the released records or prompts.

Decoding and sampling. The original profile is `eval_v4_compatible`. Its only explicitly passed generation parameter is `max_completion_tokens=8192`; temperature, top-p, model seed, reasoning effort, and stop sequences are unspecified and use service defaults. The evaluator records one saved response per model–episode combination, not repeated samples with an estimated stochastic variance. Transient request errors may be retried with the same parameters; there is no answer-quality-based resampling. Invalid-answer retry records, other protocols, and models outside the main-table roster are excluded

Table 6. **Requested API identifiers and returned model identifiers.** Parentheses count the 2,592 responses per model associated with each returned string. These are recorded service identifiers, not verified checkpoint hashes.

Requested model	Returned model strings (response count)	Endpoint
`gpt-5.6-sol`	gpt-5.6-sol-2026-07-09 (2489); gpt-5.6-sol-DataZone (22); gpt-5.6-sol (81)	A
`claude-opus-5`	claude-opus-5 (605); global.anthropic.claude-opus-5 (1987)	A
`claude-sonnet-5`	global.anthropic.claude-sonnet-5 (1556); claude-sonnet-5 (1036)	A
`gemini-2.5-pro`	gemini-2.5-pro (2592)	A
`gpt-4o`	gpt-4o (1127); gpt-4o-2024-11-20 (1464); gpt-4o-2024-08-06 (1)	A
`intern-s1`	Intern-S1 (2592)	B
`intern-s1-pro`	Intern-S1-Pro (2592)	B
`Qwen/Qwen3-VL-8B-Thinking`	Qwen/Qwen3-VL-8B-Thinking (2592)	A
`glm-5v-turbo`	glm-5v-turbo (2592)	A
`gemini-2.5-flash`	gemini-2.5-flash (2592)	A
`glm-4.6v`	glm-4.6v (2592)	A
`kimi/kimi-k2.5`	kimi-k2.5 (2592)	A
`Qwen/Qwen3-VL-32B-Thinking`	Qwen/Qwen3-VL-32B-Thinking (2592)	A
`intern-s1-mini`	intern-s1-mini (2592)	B

Table 7. **Access dates, invalid outputs, and token accounting.** Dates are UTC month–day in 2026. Invalid and Length are counts for choice/judgment (C/J); Length is the provider's finish reason and need not imply an invalid answer. Prompt and Completion are mean provider-reported tokens per ordinary episode.

		Invalid		Length		Mean tokens	
Model	UTC dates	C	J	C	J	Prompt	Completion
GPT-5.6-Sol	09-07–09-14	0	0	0	0	8628	433
Claude Opus 5	09-10–09-14	0	0	1	0	10922	592
Claude Sonnet 5	09-28–10-01	25	0	26	0	10956	738
Gemini 2.5 Pro	09-07–09-10	5	0	191	0	4404	3093
GPT-4o	09-07–09-09	0	0	0	0	7021	65
Intern-S1	09-29–10-01	11	0	15	0	20181	979
Intern-S1-Pro	09-29–09-30	18	2	71	2	7953	1381
Qwen3-VL-8B-Thinking	09-25–09-27	2	1	0	0	7893	3620
GLM-5V-Turbo	09-10–09-16	1	3	0	0	9624	2147
Gemini 2.5 Flash	09-25–09-26	10	0	72	0	4404	2271
GLM-4.6V	09-17–09-18	0	0	0	0	9574	1061
Kimi K2.5	09-16–09-16	0	27	1	0	10162	311
Qwen3-VL-32B-Thinking	09-14–09-17	0	0	0	0	7893	3474
Intern-S1-Mini	09-29–09-29	6	1	4	0	20181	899

All 36,288 MLLM records contain prompt/completion/total token fields. Tokenization and hidden reasoning accounting vary by service; these values are recorded usage, not a common measure of computation or a price estimate. Random uses no API and is omitted.

from every reported simulation statistic. Random uses the recorded seed-0 pseudorandom run, samples uniformly from the two numbered options, and makes no model API request.

Versions and accounting limits. Access dates, invalid counts, length finish counts, and mean token usage are listed in Table 7; full counts and endpoint, fingerprint, service-tier, and returned-name distributions are retained in `simulation_inference.json`. For example, GPT-4o has three returned identifiers, including both dated August and November 2024 strings, while GPT-5.6-Sol has a dated identifier, its unsuffixed alias, and a DataZone alias. The pooled scores retain these variations rather than retrospectively selecting a returned version. Identifier changes can reflect aliases or routing; the records do not determine whether weights changed. Checkpoint hashes, quantization, upstream revision histories, and the evaluation-time client environment are not pinned by these records. Names associated with publicly distributed models are still evaluated as API services; local checkpoint equivalence and a common internal reasoning configuration cannot be asserted. Provider-reported token counts include service-specific image and, where applicable, hidden reasoning accounting. They do not provide a uniform computation measure. Actual invoices or historical billing rates are absent, so monetary cost is not inferred.

E. Metrics and statistical analysis

The unit of evaluation is an episode; the 27,648 steps are not treated as independent samples. All intervals and paired comparisons cluster on the nuisance-pair identifier.

Figure 1 is a conceptual illustration, not a quantitative result plot. The object-recognition examples and mirror im-

age motivate the self-recognition question; they are not evidence inputs in VISUOSELF. The arm diagrams summarize the two decision formats, and the displayed responses are schematic rather than verbatim model transcripts. In judgment, the single visible arm is indexed locally; its label does not establish whether it is self. Model performance is quantified in Table 4 and Figures 3–5.

For scene-level rates reported in the main table and text, the estimator is the equal-weight mean of the three difficulty levels within each scene, equivalent to pooling 432 episodes per format or 216 class-specific judgment episodes. Invalid responses remain in the denominator and count as incorrect. High self recall together with low non-self recall indicates a permissive self-attribution criterion; high self recall alone does not establish discrimination.

For Figure 4, we compute accuracy within each scene from all 144 episodes at a given difficulty and format. Self and non-self recall use the 72 judgment episodes of the corresponding class in that scene. The plotted estimator is the equal-weight mean of the three scene-specific rates, equivalent here to pooling 432 episodes for each accuracy point or 216 class-specific episodes for each recall point. Invalid responses remain in their respective denominators and count as incorrect.

For the simulation difficulty plot, each bootstrap replicate independently resamples 72 complete nuisance pairs with replacement within each scene. Both conditions of each sampled pair are retained, and the same sampled pair multiplicities are shared across all four models, three difficulty levels, and two formats. We then recompute the scene means and their equal-weight average. We use 5,000 replicates with random seed 2026 and take the 2.5th and 97.5th percentiles as the interval endpoints. These are marginal, pointwise 95% intervals, not simultaneous bands or paired significance tests. The 50% reference is the expected accuracy and class-conditional recall of uniform binary prediction, not a fitted curve or the realized Random baseline.

Audit intervals use the same scene-stratified cluster design, now shared across all fourteen models. Within each scene we resample 72 base nuisance pairs and retain both conditions, both formats, and all three difficulties. Each replicate recomputes the original accuracy, APR, and LAA from the resampled episode counts; APR uses the resampled initially correct count as its denominator. The analysis uses 5,000 draws, seed 2026, and marginal 2.5th/97.5th percentiles. Undefined conditional rates in replicates with no initially correct answers are omitted, and their number is recorded in the machine-readable output; this affects sparse class-specific APR estimates, not the all-episode LAA denominator. Table 5 reports LAA intervals. The accompanying analysis also records intervals for every scene–difficulty–format audit cell. These are descriptive intervals, not multiple-comparison-adjusted significance claims.

Complete ordinary statistics and paired transfer. The same draws are also used for all fifteen rows of the original simulation table, including Random, and for PTC. For pair–level unit $v = (s, u, l)$, let c_{v1}, c_{v2} denote the two choice correctness indicators and j_v^+, j_v^- the two judgment indicators. With $N = 648$ units,

$$\begin{aligned} \mathrm{PTC} &= \frac{1}{2N} \sum_v (c_{v1} + c_{v2}) j_v^+ j_v^-, \\ \mathrm{PTC}_4 &= \frac{1}{N} \sum_v c_{v1} c_{v2} j_v^+ j_v^-. \end{aligned} \tag{8}$$

The first estimator averages 1,296 overlapping triplets; the second averages 648 four-response sets. Neither treats the repeated judgments or the three difficulty levels as independent observations. Conditional transfer divides the number of successful triplets by the number of correct choices. Each replicate recomputes these ratios from complete resampled base pairs. Triplet PTC is bounded by the fraction of judgment pairs with both classes correct; accepting every arm as self gives PTC zero regardless of choice accuracy. Theoretical uniform independent predictions give $1/8$ PTC and $1/16$ PTC_4; the empirical Random row is reported separately.

Paired format tests. For each of the fourteen MLLMs we test the ordinary accuracy difference $\widehat{\Delta} = \widehat{A}_C - \widehat{A}_J$. The same cluster draws yield paired replicates Δ_r^* and percentile intervals. An approximate two-sided null-centered bootstrap test uses

$$p = \frac{1 + \sum_{r=1}^{R} \mathbb{1}[|\Delta_r^* - \widehat{\Delta}| \geq |\widehat{\Delta}|]}{R + 1}, \quad R = 5{,}000. \tag{9}$$

Holm's step-down adjustment applies to the family of fourteen format contrasts, with $\alpha = 0.05$. It does not turn the marginal intervals into simultaneous confidence bands. Table 9 gives every contrast, interval, raw p, and adjusted p. The minimum Monte Carlo resolution is $1/5{,}001$; these are approximate sampling tests for the frozen episode population, not randomized causal effects of decision format.

Response bias and extreme rates. Accuracy and class/position recalls retain all original episodes; invalid answers receive zero correctness credit. Semantic self attribution and choice-position prediction fractions also divide by all episodes: invalid responses enter neither prediction category. Signal-detection diagnostics use only valid judgment responses because an invalid output is neither a self nor a non-self decision. If h, f count self predictions on valid self and non-self episodes, respectively, and n_+, n_- are their valid totals, we use $H = (h + 0.5)/(n_+ + 1)$ and $F = (f + 0.5)/(n_- + 1)$, then

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(H) + \Phi^{-1}(F)]. \tag{10}$$

The half-count correction keeps the estimates finite at zero/one rates; negative c denotes a permissive self criterion. These valid-only diagnostics cannot replace the all-episode accuracy or recalls [19].

Nuisance slices and worst groups. We pool scenes and levels within each robot, environment, and view slice. Within a format, robot and environment groups have 432 episodes each, view groups 648, and each of the 18 joint combinations 72. Worst-group accuracy is the minimum of the eighteen observed joint-group rates, without a claim of a significant difference between selected groups. Full group counts and marginal intervals are retained in the machine-readable analysis; the same scene-stratified base-pair resampling is used throughout. Undefined conditional-rate replicates are omitted and counted explicitly.

F. Complete simulation results

All tables in this section are generated from the same 38,880 original responses: fourteen MLLMs and Random, each with 2,592 episodes. Figure 10 summarizes all 270 scene–difficulty–format accuracy cells; Tables 13–15 give their values and intervals. Tables 16–18 give self/non-self recalls in all 135 judgment cells. PTC cells, format contrasts, position/semantic bias, and nuisance slices are reported in Tables 10, 9, 11, and 12. Table 8 reports the overall PTC, PTC_4, conditional transfer, and ordinary accuracy intervals underlying the main-paper outcome plot.

Nuisance variation does not eliminate the principal asymmetry. GPT-5.6-Sol's lowest joint-group choice accuracy is 81.9%, while its lowest joint-group judgment accuracy is 50.0%. Gemini 2.5 Pro's corresponding minima are 52.8% and 68.1%, and Claude Opus 5's are 86.1% and 56.9%. These descriptive minima complement the marginal slices; the small joint groups do not identify individual nuisance factors as causal sources of errors.

Complete audit coverage and scene-specific logic-adjusted accuracy are given in Appendix G; all 252 model–scene–difficulty–format audit cells are additionally recorded in the generated analysis artifacts. Physical results remain separate in Appendix J. The ordinary companion `simulation_results.json`/CSV contains 1,260 summary groups with counts and intervals, and `simulation_ptc.csv` contains 240 transfer groups. These include all marginal and joint nuisance slices, not just the worst groups printed here. The analysis validates source hashes against the full-input verification receipt and separately validates Random's labels, parsing, and shared input hashes. No model output is regenerated.

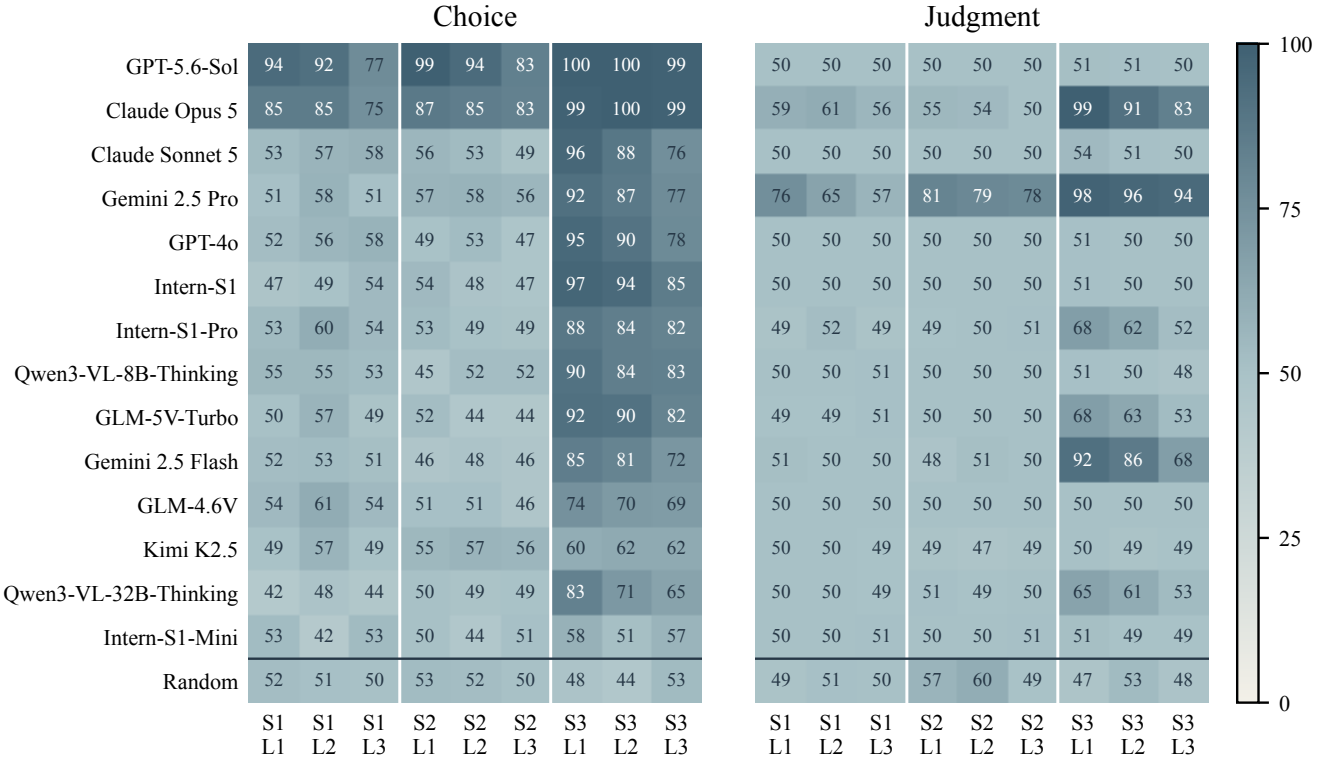


Figure 10. **All original simulation accuracy cells.** Columns index scene and difficulty; both panels use the same 0–100% scale. Numbers are rounded to whole percentages for display; the tables provide one-decimal values and intervals. Random is separated below the fourteen MLLMs. Each cell contains 144 original responses, with invalid outputs incorrect.

Table 8. **Cross-format consistency and original-accuracy intervals.** PTC requires one correct choice and both correct isolated judgments of matched behaviors. PTC_4 additionally requires choice to succeed at both target positions. Transfer is the probability that both judgments are correct, conditional on a correct choice.

Model	Choice [95% CI]	Judgment [95% CI]	PTC [95% CI]	PTC_4	Transfer
GPT-5.6-Sol	93.1 [91.5, 94.5]	50.2 [50.0, 50.5]	0.5 [0.0, 1.1]	0.5	0.5
Claude Opus 5	88.6 [86.9, 90.2]	67.6 [66.1, 69.1]	34.0 [31.3, 36.7]	33.0	38.4
Claude Sonnet 5	65.1 [62.6, 67.6]	50.5 [50.2, 51.0]	1.0 [0.3, 1.9]	0.9	1.5
Gemini 2.5 Pro	65.0 [62.6, 67.4]	80.5 [78.2, 82.6]	43.9 [40.8, 47.0]	31.6	67.6
GPT-4o	64.0 [61.8, 66.2]	50.1 [50.0, 50.2]	0.2 [0.0, 0.5]	0.2	0.2
Intern-S1	64.0 [61.9, 66.0]	50.1 [50.0, 50.2]	0.2 [0.0, 0.5]	0.2	0.2
Intern-S1-Pro	63.6 [61.1, 66.2]	53.5 [52.2, 55.0]	9.5 [7.6, 11.5]	7.4	14.9
Qwen3-VL-8B-Thinking	63.3 [60.8, 65.7]	49.9 [49.4, 50.4]	0.5 [0.1, 1.2]	0.5	0.9
GLM-5V-Turbo	62.2 [59.8, 64.7]	53.7 [52.9, 54.6]	6.7 [5.1, 8.4]	5.9	10.8
Gemini 2.5 Flash	59.3 [57.2, 61.5]	60.7 [59.4, 62.0]	18.3 [16.2, 20.4]	14.5	30.8
GLM-4.6V	58.9 [56.9, 60.9]	50.0 [50.0, 50.0]	0.0 [0.0, 0.0]	0.0	0.0
Kimi K2.5	56.3 [53.6, 59.0]	49.1 [48.5, 49.6]	0.1 [0.0, 0.2]	0.0	0.1
Qwen3-VL-32B-Thinking	55.7 [53.5, 57.9]	53.2 [52.2, 54.3]	5.2 [3.7, 6.9]	3.9	9.3
Intern-S1-Mini	51.0 [48.5, 53.5]	50.2 [49.7, 50.6]	0.5 [0.1, 1.1]	0.3	1.1
Random	50.3 [47.6, 53.2]	51.6 [48.7, 54.6]	13.3 [11.0, 15.7]	6.6	26.5

All rates are percentages. Per model: 1,296 choices, 1,296 judgments, 1,296 overlapping PTC triplets, and 648 four-response sets. Intervals resample 216 base nuisance pairs, retaining all levels and formats. Random is the recorded seed-0 run; its theoretical PTC/PTC_4 expectations are 12.5%/6.25%.

Table 9. **Paired format differences.** Δ is choice minus judgment accuracy in percentage points. Intervals are marginal 95% paired cluster-bootstrap intervals. p is the two-sided zero-centered bootstrap test; p_{Holm} controls the family of fourteen model-wise comparisons.

Model	Δ	[95% CI]	p	p_{Holm}
GPT-5.6-Sol	+42.8	[+41.2, +44.3]	0.0002	0.0028
Claude Opus 5	+21.0	[+18.9, +23.1]	0.0002	0.0028
Claude Sonnet 5	+14.6	[+12.0, +17.1]	0.0002	0.0028
Gemini 2.5 Pro	-15.5	[-18.8, -12.2]	0.0002	0.0028
GPT-4o	+14.0	[+11.7, +16.1]	0.0002	0.0028
Intern-S1	+13.9	[+11.8, +15.9]	0.0002	0.0028
Intern-S1-Pro	+10.0	[+7.3, +12.9]	0.0002	0.0028
Qwen3-VL-8B-Thinking	+13.3	[+10.9, +15.7]	0.0002	0.0028
GLM-5V-Turbo	+8.5	[+5.9, +11.0]	0.0002	0.0028
Gemini 2.5 Flash	-1.4	[-3.6, +1.0]	0.2637	0.5275
GLM-4.6V	+8.9	[+6.9, +10.9]	0.0002	0.0028
Kimi K2.5	+7.3	[+4.5, +10.0]	0.0002	0.0028
Qwen3-VL-32B-Thinking	+2.5	[+0.2, +4.9]	0.0490	0.1470
Intern-S1-Mini	+0.8	[-1.5, +3.3]	0.5105	0.5275

The same $5{,}000$ scene-stratified pair draws preserve both formats and all difficulties. The smallest attainable Monte Carlo p is $1/5{,}001$; displayed 0.0002 values are at that resolution. These tests concern the frozen API snapshot and do not establish invariance to prompt or backend changes.

Table 10. **PTC by scene and difficulty.** Each cell uses 144 overlapping triplets from 72 behavior pairs. Values and bracketed marginal 95% intervals are percentages.

	Scene 1			Scene 2			Scene 3		
Model	L1	L2	L3	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	2.8 [0.0, 6.9]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]
Claude Opus 5	16.7 [8.3, 25.0]	19.4 [11.1, 28.5]	10.4 [4.2, 17.4]	7.6 [2.8, 13.2]	7.6 [2.8, 13.9]	0.0 [0.0, 0.0]	97.2 [93.8, 100.0]	81.9 [72.2, 90.3]	65.3 [54.2, 75.7]
Claude Sonnet 5	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	8.3 [2.8, 15.3]	0.7 [0.0, 2.1]	0.0 [0.0, 0.0]
Gemini 2.5 Pro	27.1 [19.4, 35.4]	20.8 [13.2, 29.2]	10.4 [4.9, 16.7]	35.4 [27.1, 44.4]	35.4 [27.1, 44.4]	29.9 [21.5, 38.2]	88.2 [81.9, 93.8]	80.6 [72.9, 87.5]	67.4 [58.3, 76.4]
GPT-4o	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1-Pro	4.2 [0.7, 9.0]	6.2 [2.1, 11.1]	2.8 [0.0, 6.2]	0.7 [0.0, 2.1]	0.7 [0.0, 2.1]	4.9 [0.7, 10.4]	33.3 [22.9, 43.8]	24.3 [15.3, 34.0]	8.3 [2.8, 14.6]
Qwen3-VL-8B-Thinking	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.7 [0.0, 2.1]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]
GLM-5V-Turbo	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.7 [0.0, 2.1]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	32.6 [22.9, 43.1]	22.9 [13.9, 32.6]	4.2 [0.7, 9.0]
Gemini 2.5 Flash	1.4 [0.0, 3.5]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.7 [0.0, 2.1]	3.5 [0.7, 7.6]	0.0 [0.0, 0.0]	73.6 [64.6, 81.9]	59.7 [50.0, 69.4]	25.7 [16.7, 34.7]
GLM-4.6V	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Kimi K2.5	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.7 [0.0, 2.1]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Qwen3-VL-32B-Thinking	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	24.3 [15.3, 34.0]	16.0 [9.0, 23.6]	4.9 [1.4, 9.7]
Intern-S1-Mini	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.7 [0.0, 2.1]	1.4 [0.0, 3.5]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Random	16.0 [9.0, 23.6]	16.0 [9.0, 23.6]	11.1 [4.9, 18.1]	14.6 [8.3, 21.5]	19.4 [11.8, 27.8]	11.1 [6.2, 16.7]	12.5 [6.2, 19.4]	10.4 [5.6, 16.0]	9.0 [4.2, 14.6]

Table 11. **Prediction bias and validity.** Choice recalls R_1, R_2 condition on the true target position; P_1 is the fraction of all choices predicting position 1. Judgment Self is the fraction of all judgments predicting the semantic self option. Invalid is the percentage of invalid outputs. d' and criterion c use only valid binary judgments with half-count correction.

	Choice				Judgment			
Model	R_1	R_2	P_1	Invalid	Self	Invalid	d'	c
GPT-5.6-Sol	93.7	92.4	50.6	0.0	99.8	0.0	0.62	-2.86
Claude Opus 5	85.8	91.4	47.2	0.0	82.4	0.0	2.79	-1.77
Claude Sonnet 5	54.2	76.1	38.3	1.9	99.5	0.0	0.90	-2.72
Gemini 2.5 Pro	67.6	62.3	52.5	0.4	53.6	0.0	1.73	-0.13
GPT-4o	45.5	82.6	31.5	0.0	99.9	0.0	0.33	-3.00
Intern-S1	78.1	49.8	63.7	0.8	99.9	0.0	0.33	-3.00
Intern-S1-Pro	62.0	65.1	47.6	1.4	90.9	0.2	0.47	-1.38
Qwen3-VL-8B-Thinking	69.0	57.6	55.6	0.2	99.1	0.1	-0.06	-2.36
GLM-5V-Turbo	66.2	58.2	53.9	0.1	95.8	0.2	1.41	-2.13
Gemini 2.5 Flash	87.7	31.0	78.1	0.8	88.2	0.0	1.52	-1.51
GLM-4.6V	87.8	29.9	78.9	0.0	100.0	0.0	0.00	-3.17
Kimi K2.5	65.6	47.1	59.3	0.0	97.8	2.1	0.34	-2.99
Qwen3-VL-32B-Thinking	81.2	30.2	75.5	0.0	96.5	0.0	1.18	-2.08
Intern-S1-Mini	70.7	31.3	69.6	0.5	99.2	0.1	0.22	-2.44
Random	50.5	50.2	50.2	0.0	51.6	0.0	0.08	-0.04

Rates are percentages; d' and c are dimensionless. Invalid answers remain incorrect in all accuracy/recall denominators; they enter neither semantic prediction category. Signal-detection statistics are conditional diagnostics, not replacements for all-episode performance.

Table 12. **Nuisance-factor slices and worst-group accuracy.** C/J denote choice/judgment. Marginal slices pool scenes and difficulties; Worst is the minimum across the 18 joint robot–environment–camera groups within a format. All values are percentages.

		Robot			Environment			View		
Model		Cobotta	Panda	UR5	Assembly	Inspection	Lab	Front	Overhead	Worst
GPT-5.6-Sol	C	92.8	95.4	91.0	93.3	93.5	92.4	94.9	91.2	81.9
	J	50.5	50.0	50.2	50.2	50.5	50.0	50.2	50.3	50.0
Claude Opus 5	C	88.4	89.1	88.2	88.4	90.0	87.3	87.5	89.7	86.1
	J	63.7	65.0	74.1	66.9	68.3	67.6	66.8	68.4	56.9
Claude Sonnet 5	C	66.2	64.8	64.4	65.0	65.5	64.8	63.9	66.4	56.9
	J	50.2	50.2	51.2	50.9	50.0	50.7	50.8	50.3	50.0
Gemini 2.5 Pro	C	61.8	66.2	66.9	65.5	64.6	64.8	68.2	61.7	52.8
	J	78.2	82.9	80.3	79.4	80.6	81.5	81.0	79.9	68.1
GPT-4o	C	63.0	64.8	64.4	62.5	67.4	62.3	63.1	65.0	54.2
	J	50.2	50.0	50.0	50.2	50.0	50.0	50.0	50.2	50.0
Intern-S1	C	60.9	65.0	66.0	62.5	65.5	63.9	62.7	65.3	56.9
	J	50.0	50.2	50.0	50.2	50.0	50.0	50.0	50.2	50.0
Intern-S1-Pro	C	62.7	65.5	62.5	64.8	65.3	60.6	61.7	65.4	50.0
	J	53.5	53.9	53.2	53.5	53.7	53.5	52.8	54.3	48.6
Qwen3-VL-8B-Thinking	C	59.3	64.8	65.7	62.0	65.3	62.5	63.6	63.0	47.2
	J	49.8	50.0	50.0	49.5	50.0	50.2	50.2	49.7	48.6
GLM-5V-Turbo	C	63.0	63.0	60.6	62.7	59.7	64.1	62.2	62.2	51.4
	J	53.7	52.8	54.6	53.7	53.9	53.5	52.5	54.9	50.0
Gemini 2.5 Flash	C	56.2	60.4	61.3	57.4	61.8	58.8	55.2	63.4	47.2
	J	57.4	62.7	62.0	60.2	61.1	60.9	60.0	61.4	51.4
GLM-4.6V	C	57.6	60.4	58.6	60.6	58.8	57.2	59.0	58.8	52.8
	J	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0
Kimi K2.5	C	55.3	53.5	60.2	53.0	55.6	60.4	56.3	56.3	45.8
	J	49.5	48.6	49.1	49.5	49.1	48.6	49.2	48.9	45.8
Qwen3-VL-32B-Thinking	C	54.9	59.7	52.5	54.2	54.9	58.1	56.5	54.9	47.2
	J	52.1	52.8	54.9	54.6	52.1	53.0	52.5	54.0	50.0
Intern-S1-Mini	C	48.8	50.2	53.9	51.9	48.4	52.8	52.3	49.7	40.3
	J	50.0	50.0	50.5	50.0	50.0	50.5	50.0	50.3	48.6
Random	C	48.8	50.5	51.6	48.6	50.9	51.4	47.4	53.2	43.1
	J	51.4	50.0	53.5	48.8	51.9	54.2	51.2	52.0	38.9

Robot/environment marginal groups contain 432 episodes per format, view groups 648, and joint groups 72. The JSON/CSV companion records counts and intervals for every marginal and joint group. Worst groups are selected descriptively, not claimed to differ significantly.

Table 13. **Complete Scene 1 answer accuracy.** Each entry uses 144 ordinary episodes. Brackets give marginal 95% nuisance-pair bootstrap intervals; all values are percentages.

	Choice			Judgment		
Model	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	93.8 [88.9, 97.2]	91.7 [86.1, 96.5]	77.1 [69.4, 84.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Claude Opus 5	85.4 [79.9, 91.0]	85.4 [79.9, 91.0]	75.0 [68.1, 81.9]	59.0 [54.9, 63.9]	61.1 [56.2, 66.0]	55.6 [52.1, 59.0]
Claude Sonnet 5	53.5 [45.8, 61.1]	56.9 [49.3, 64.6]	58.3 [50.7, 66.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Gemini 2.5 Pro	50.7 [43.1, 58.3]	57.6 [50.0, 65.3]	50.7 [43.8, 57.6]	75.7 [69.4, 81.9]	65.3 [58.3, 71.5]	56.9 [51.4, 62.5]
GPT-4o	52.1 [45.1, 59.0]	55.6 [47.9, 63.2]	57.6 [51.4, 63.9]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Intern-S1	47.2 [39.6, 54.2]	49.3 [41.7, 56.2]	54.2 [47.2, 60.4]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Intern-S1-Pro	53.5 [45.8, 61.1]	60.4 [53.5, 67.4]	54.2 [46.5, 61.8]	48.6 [44.4, 52.8]	52.1 [47.9, 56.9]	48.6 [43.8, 52.8]
Qwen3-VL-8B-Thinking	54.9 [47.9, 61.8]	54.9 [47.2, 62.5]	52.8 [44.4, 61.1]	50.0 [47.2, 52.8]	50.0 [50.0, 50.0]	50.7 [50.0, 52.1]
GLM-5V-Turbo	50.0 [42.4, 57.6]	56.9 [50.0, 63.9]	48.6 [41.7, 55.6]	49.3 [47.9, 50.0]	49.3 [47.9, 50.0]	50.7 [50.0, 52.1]
Gemini 2.5 Flash	52.1 [47.9, 56.9]	52.8 [47.9, 57.6]	51.4 [46.5, 56.2]	51.4 [50.0, 53.5]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
GLM-4.6V	54.2 [48.6, 59.7]	61.1 [55.6, 67.4]	54.2 [48.6, 59.7]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Kimi K2.5	49.3 [42.4, 56.2]	56.9 [50.7, 63.2]	49.3 [41.7, 56.2]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	49.3 [47.9, 50.0]
Qwen3-VL-32B-Thinking	42.4 [35.4, 49.3]	47.9 [41.0, 54.9]	44.4 [38.2, 50.7]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	49.3 [47.9, 50.0]
Intern-S1-Mini	52.8 [45.1, 60.4]	41.7 [34.7, 48.6]	53.5 [45.8, 61.1]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.7 [50.0, 52.1]
Random	52.1 [43.8, 60.4]	50.7 [43.1, 59.0]	50.0 [41.7, 58.3]	48.6 [39.6, 57.6]	50.7 [42.4, 59.0]	50.0 [41.0, 59.0]

Table 14. **Complete Scene 2 answer accuracy.** Each entry uses 144 ordinary episodes. Brackets give marginal 95% nuisance-pair bootstrap intervals; all values are percentages.

	Choice			Judgment		
Model	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	99.3 [97.9, 100.0]	93.8 [89.6, 97.2]	83.3 [76.4, 89.6]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Claude Opus 5	86.8 [81.2, 91.7]	84.7 [79.2, 90.3]	82.6 [76.4, 88.2]	54.9 [52.1, 58.3]	54.2 [51.4, 57.6]	50.0 [50.0, 50.0]
Claude Sonnet 5	55.6 [47.2, 63.9]	53.5 [45.8, 61.1]	48.6 [41.7, 54.9]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Gemini 2.5 Pro	56.9 [49.3, 65.3]	57.6 [49.3, 66.0]	55.6 [48.6, 63.2]	81.2 [75.0, 86.8]	79.2 [73.6, 84.7]	77.8 [72.2, 83.3]
GPT-4o	49.3 [43.1, 56.2]	52.8 [47.2, 58.3]	46.5 [41.0, 52.8]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Intern-S1	54.2 [48.6, 59.7]	47.9 [41.0, 54.9]	46.5 [40.3, 52.8]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Intern-S1-Pro	52.8 [45.1, 61.1]	49.3 [41.7, 56.2]	48.6 [40.3, 56.9]	49.3 [46.5, 52.1]	50.0 [47.9, 52.1]	50.7 [47.2, 54.2]
Qwen3-VL-8B-Thinking	45.1 [37.5, 52.8]	52.1 [45.1, 59.0]	52.1 [45.8, 58.3]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
GLM-5V-Turbo	52.1 [45.8, 58.3]	43.8 [36.8, 50.7]	44.4 [36.8, 52.1]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Gemini 2.5 Flash	45.8 [40.3, 51.4]	47.9 [41.0, 54.2]	45.8 [40.3, 51.4]	47.9 [45.1, 50.7]	51.4 [47.9, 54.9]	50.0 [47.9, 52.1]
GLM-4.6V	50.7 [44.4, 56.9]	50.7 [45.1, 56.2]	45.8 [42.4, 49.3]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Kimi K2.5	54.9 [47.9, 61.8]	56.9 [49.3, 63.9]	55.6 [49.3, 62.5]	48.6 [46.5, 50.0]	47.2 [44.4, 49.3]	48.6 [46.5, 50.0]
Qwen3-VL-32B-Thinking	50.0 [43.8, 56.2]	48.6 [43.1, 54.2]	49.3 [44.4, 53.5]	50.7 [50.0, 52.1]	49.3 [47.9, 50.0]	50.0 [50.0, 50.0]
Intern-S1-Mini	50.0 [43.1, 56.9]	43.8 [36.8, 51.4]	50.7 [43.8, 57.6]	50.0 [50.0, 50.0]	50.0 [47.9, 52.1]	50.7 [48.6, 52.8]
Random	52.8 [44.4, 60.4]	52.1 [44.4, 60.4]	50.0 [42.4, 57.6]	56.9 [49.3, 64.6]	60.4 [52.1, 68.8]	49.3 [41.0, 57.6]

Table 15. **Complete Scene 3 answer accuracy.** Each entry uses 144 ordinary episodes. Brackets give marginal 95% nuisance-pair bootstrap intervals; all values are percentages.

	Choice			Judgment		
Model	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	98.6 [96.5, 100.0]	51.4 [50.0, 53.5]	50.7 [50.0, 52.1]	50.0 [50.0, 50.0]
Claude Opus 5	98.6 [96.5, 100.0]	100.0 [100.0, 100.0]	98.6 [96.5, 100.0]	99.3 [97.9, 100.0]	91.0 [86.1, 95.1]	83.3 [77.8, 88.9]
Claude Sonnet 5	95.8 [92.4, 98.6]	88.2 [82.6, 93.1]	75.7 [68.8, 81.9]	54.2 [51.4, 57.6]	50.7 [50.0, 52.1]	50.0 [50.0, 50.0]
Gemini 2.5 Pro	91.7 [86.8, 95.8]	86.8 [81.2, 91.7]	77.1 [69.4, 84.0]	97.9 [95.1, 100.0]	95.8 [92.4, 98.6]	94.4 [90.3, 97.9]
GPT-4o	95.1 [91.7, 97.9]	89.6 [84.7, 94.4]	77.8 [71.5, 83.3]	50.7 [50.0, 52.1]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Intern-S1	96.5 [93.1, 99.3]	94.4 [91.0, 97.9]	85.4 [78.5, 91.7]	50.7 [50.0, 52.1]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Intern-S1-Pro	87.5 [81.9, 92.4]	84.0 [78.5, 89.6]	81.9 [76.4, 87.5]	68.1 [62.5, 73.6]	62.5 [56.9, 68.1]	52.1 [47.2, 56.9]
Qwen3-VL-8B-Thinking	90.3 [84.7, 95.1]	84.0 [78.5, 89.6]	83.3 [77.8, 88.9]	50.7 [50.0, 52.1]	50.0 [47.9, 52.1]	47.9 [45.1, 50.0]
GLM-5V-Turbo	91.7 [86.8, 95.8]	90.3 [84.7, 95.1]	81.9 [75.7, 87.5]	68.1 [62.5, 73.6]	63.2 [58.3, 68.1]	52.8 [50.7, 55.6]
Gemini 2.5 Flash	84.7 [79.2, 89.6]	81.2 [75.7, 86.8]	72.2 [65.3, 79.2]	91.7 [87.5, 95.8]	86.1 [80.6, 91.0]	68.1 [62.5, 73.6]
GLM-4.6V	74.3 [67.4, 80.6]	70.1 [63.9, 76.4]	68.8 [62.5, 75.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Kimi K2.5	60.4 [52.1, 68.8]	61.8 [54.2, 69.4]	61.8 [54.2, 69.4]	50.0 [47.9, 52.1]	48.6 [46.5, 50.0]	49.3 [47.9, 50.0]
Qwen3-VL-32B-Thinking	82.6 [76.4, 88.9]	70.8 [63.2, 77.8]	65.3 [57.6, 72.9]	65.3 [59.7, 70.8]	61.1 [56.2, 66.0]	53.5 [50.7, 56.9]
Intern-S1-Mini	58.3 [51.4, 65.3]	51.4 [44.4, 58.3]	56.9 [48.6, 65.3]	51.4 [50.0, 53.5]	49.3 [47.9, 50.0]	49.3 [47.9, 50.0]
Random	47.9 [41.0, 55.6]	43.8 [36.1, 51.4]	53.5 [45.1, 61.8]	47.2 [38.9, 55.6]	53.5 [45.8, 61.1]	47.9 [40.3, 55.6]

Table 16. **Complete Scene 1 judgment recalls.** Each entry uses 72 episodes of its class; invalid answers count as incorrect. Brackets give marginal 95% nuisance-pair bootstrap intervals; all values are percentages.

	Self recall			Non-self recall		
Model	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Claude Opus 5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	18.1 [9.7, 27.8]	22.2 [12.5, 31.9]	11.1 [4.2, 18.1]
Claude Sonnet 5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Gemini 2.5 Pro	97.2 [93.1, 100.0]	90.3 [83.3, 95.8]	91.7 [84.7, 97.2]	54.2 [43.1, 65.3]	40.3 [29.2, 51.4]	22.2 [12.5, 31.9]
GPT-4o	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1-Pro	90.3 [83.3, 95.8]	91.7 [84.7, 97.2]	90.3 [83.3, 95.8]	6.9 [1.4, 12.5]	12.5 [5.6, 20.8]	6.9 [1.4, 13.9]
Qwen3-VL-8B-Thinking	97.2 [93.1, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	2.8 [0.0, 6.9]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]
GLM-5V-Turbo	98.6 [95.8, 100.0]	98.6 [95.8, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]
Gemini 2.5 Flash	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	2.8 [0.0, 6.9]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
GLM-4.6V	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Kimi K2.5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Qwen3-VL-32B-Thinking	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1-Mini	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]
Random	50.0 [38.9, 61.1]	50.0 [38.9, 61.1]	47.2 [36.1, 58.3]	47.2 [36.1, 58.3]	51.4 [40.3, 62.5]	52.8 [41.7, 63.9]

Table 17. **Complete Scene 2 judgment recalls.** Each entry uses 72 episodes of its class; invalid answers count as incorrect. Brackets give marginal 95% nuisance-pair bootstrap intervals; all values are percentages.

	Self recall			Non-self recall		
Model	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Claude Opus 5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	9.7 [4.2, 16.7]	8.3 [2.8, 15.3]	0.0 [0.0, 0.0]
Claude Sonnet 5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Gemini 2.5 Pro	63.9 [52.8, 75.0]	59.7 [48.6, 70.8]	65.3 [54.2, 76.4]	98.6 [95.8, 100.0]	98.6 [95.8, 100.0]	90.3 [83.3, 97.2]
GPT-4o	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1-Pro	95.8 [90.3, 100.0]	98.6 [95.8, 100.0]	95.8 [90.3, 100.0]	2.8 [0.0, 6.9]	1.4 [0.0, 4.2]	5.6 [1.4, 11.1]
Qwen3-VL-8B-Thinking	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
GLM-5V-Turbo	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Gemini 2.5 Flash	94.4 [88.9, 98.6]	97.2 [93.1, 100.0]	98.6 [95.8, 100.0]	1.4 [0.0, 4.2]	5.6 [1.4, 11.1]	1.4 [0.0, 4.2]
GLM-4.6V	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Kimi K2.5	97.2 [93.1, 100.0]	94.4 [88.9, 98.6]	97.2 [93.1, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Qwen3-VL-32B-Thinking	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	100.0 [100.0, 100.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1-Mini	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	98.6 [95.8, 100.0]	0.0 [0.0, 0.0]	1.4 [0.0, 4.2]	2.8 [0.0, 6.9]
Random	58.3 [47.2, 69.4]	66.7 [55.6, 77.8]	54.2 [43.1, 65.3]	55.6 [44.4, 66.7]	54.2 [43.1, 65.3]	44.4 [33.3, 55.6]

Table 18. **Complete Scene 3 judgment recalls.** Each entry uses 72 episodes of its class; invalid answers count as incorrect. Brackets give marginal 95% nuisance-pair bootstrap intervals; all values are percentages.

	Self recall			Non-self recall		
Model	L1	L2	L3	L1	L2	L3
GPT-5.6-Sol	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	2.8 [0.0, 6.9]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]
Claude Opus 5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	81.9 [72.2, 90.3]	66.7 [55.6, 77.8]
Claude Sonnet 5	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	8.3 [2.8, 15.3]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]
Gemini 2.5 Pro	97.2 [93.1, 100.0]	91.7 [84.7, 97.2]	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	100.0 [100.0, 100.0]	88.9 [80.6, 95.8]
GPT-4o	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Intern-S1-Pro	97.2 [93.1, 100.0]	97.2 [93.1, 100.0]	93.1 [86.1, 98.6]	38.9 [27.8, 50.0]	27.8 [18.1, 38.9]	11.1 [4.2, 19.4]
Qwen3-VL-8B-Thinking	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	95.8 [90.3, 100.0]	1.4 [0.0, 4.2]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]
GLM-5V-Turbo	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	36.1 [25.0, 47.2]	26.4 [16.7, 36.1]	5.6 [1.4, 11.1]
Gemini 2.5 Flash	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	83.3 [75.0, 91.7]	72.2 [61.1, 81.9]	36.1 [25.0, 47.2]
GLM-4.6V	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Kimi K2.5	98.6 [95.8, 100.0]	97.2 [93.1, 100.0]	98.6 [95.8, 100.0]	1.4 [0.0, 4.2]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Qwen3-VL-32B-Thinking	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	30.6 [19.4, 41.7]	22.2 [12.5, 31.9]	6.9 [1.4, 13.9]
Intern-S1-Mini	100.0 [100.0, 100.0]	98.6 [95.8, 100.0]	98.6 [95.8, 100.0]	2.8 [0.0, 6.9]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Random	51.4 [40.3, 62.5]	55.6 [44.4, 66.7]	45.8 [34.7, 56.9]	43.1 [31.9, 54.2]	51.4 [40.3, 62.5]	50.0 [38.9, 61.1]

G. Structured logic audit

Scope and provenance. The audit uses `logic_audit_v2` paired with the original `eval_v4_compatible` simulation results. Across the fourteen models in Table 4, 36,288 ordinary evaluations contain 21,765 initially correct answers; every one has exactly one audit. Audits span September 19–October 11, 2026. Earlier dataset/protocol versions, invalid-answer retries, the Random baseline, and models outside the main roster are excluded. An audit's stored original-result hash must match the exact source file, and its episode signature, input hash, preceding answer, prompt hash, and generation parameters must agree with that source and the frozen episode. All raw audit responses are rescored against applied-command traces. Full-history multimodal inputs are reconstructed for all 2,592 episodes and checked against the common input hashes used by all models. These checks validate provenance and scoring without rerunning model calls.

Follow-up context. Each audit is a new API request that reconstructs the original user input (complete commands and ordered visual evidence), includes the complete preceding assistant response, and adds the audit question as a second user message. The original system message is replaced with the following audit instruction; this is a continuation context, not an independent answer-blind evaluation:

> You are auditing a completed answer about a benign Isaac Sim scene. No real-world robot is being controlled. Reinspect the original motor-command trace and visual evidence, then return only the requested structured JSON. The existence of this audit does not indicate whether the preceding answer was correct. Do not use the preceding choice as evidence.

No answer key, applied-command ground truth, or correctness flag is supplied to the model. Although the instruction denies correctness feedback, selection of initially correct answers still limits the population measured by APR. The audit allows 8,192 completion tokens, with temperature and top-p unspecified, matching the common completion limit of the ordinary evaluation.

Scene-specific prompts and schemas. The following prompts instantiate choice with candidate indices $(1, 2)$. Judgment substitutes the singleton (1); all remaining wording is identical. For readability, the JSON schema examples below are pretty-printed; their short relation arrays are illustrative, not the required output length.

Scene 1: step correspondence.

> Audit the command-motion reasoning behind the preceding answer. For each candidate arm (1, 2), classify every one of the 8 steps in chronological order. Use "match" only when that step's visible signed motion corresponds to the current motor-command delta; otherwise use "mismatch". You must commit to one of those two values for every step. Return exactly one JSON object with no Markdown or explanation, using this schema:
>
> ```
> {"audit_type":"step_match",
>  "candidates":[{"candidate":1,
>  "step_relations":["match","mismatch"]}]}
> ```
>
> Include each candidate exactly once and exactly 8 relations per candidate. The short array in the schema is illustrative only.

For candidate k at step t, the expected relation is match exactly when its saved applied vector equals the current command vector, using `allclose` with relative tolerance 10^{-7} and absolute tolerance 10^{-9}; otherwise it is mismatch. Thus judgment requires 8 items and choice requires 16.

Scene 2: cross-cycle stability.

> Audit the command-motion reasoning behind the preceding answer. The trace contains 4 repeated command cycles of 4 steps. For each candidate arm (1, 2), infer the command-to-visible-motion mapping within every cycle and compare each cycle with cycle 1. Report "same" only if the complete mapping is identical to cycle 1; otherwise report "different". Cycle 1 is therefore "same". Return exactly one JSON object with no Markdown or explanation, using this schema:
>
> ```
> {"audit_type":"cycle_stability",
>  "candidates":[{"candidate":1,
>  "relative_to_cycle_1":
>  ["same","different"]}]}
> ```
>
> Include each candidate exactly once and exactly 4 values per candidate. The short array in the schema is illustrative only.

Each candidate's ordered four-vector sequence in a cycle is compared with its first-cycle sequence, with the same vector tolerance. All four corresponding vectors must agree for same. The stipulated first-cycle value is included in scoring: judgment has 4 items and choice 8, of which 3 and 6, respectively, concern comparisons beyond the reference cycle.

Scene 3: exact temporal lag.

> Audit the command-motion reasoning behind the preceding answer. For each candidate arm (1, 2), report its exact non-negative integer lag relative to the motor-command trace. Lag 0 means the arm follows the current command; lag L means motion at step t follows the command from step t-L, with no commanded motion during the first L steps. Return exactly one JSON object with no Markdown or explanation, using this schema:
>
> ```
> {"audit_type":"temporal_lag",
>  "candidates":[{"candidate":1,"lag":0}]}
> ```
>
> Include each candidate exactly once.

Ground truth is the unique integer lag whose zero-padded shifted command trace matches the candidate's complete applied trace. Non-identifiable lag traces raise an error in the scorer. Judgment requires one lag; choice requires two. This audit tests the exact delay, beyond simply selecting the zero-lag candidate.

Table 19. **Audit coverage, structural validity, and judgment class decomposition.** C/N_+ is completed audits / initially correct answers; Valid is the percentage with a valid structure. Item is micro accuracy over required audit items among audited responses, including invalid outputs. $R^{\text{audit}}_{\text{self/non}}$ requires both the original judgment and its audit to be correct, divided by all 648 episodes of that class.

	Choice			Judgment			Audit-adjusted recall	
Model	C/N_+	Valid	Item	C/N_+	Valid	Item	$R^{\text{audit}}_{\text{self}}$	$R^{\text{audit}}_{\text{non}}$
GPT-5.6-Sol	1206/1206	100.0	91.5	651/651	100.0	99.9	99.7	0.5
Claude Opus 5	1148/1148	98.2	88.2	876/876	97.4	94.4	98.5	27.2
Claude Sonnet 5	844/844	100.0	76.6	655/655	100.0	99.8	99.7	0.8
Gemini 2.5 Pro	842/842	98.1	75.5	1043/1043	99.8	84.9	83.0	32.9
GPT-4o	830/830	100.0	77.9	649/649	100.0	99.7	99.4	0.0
Intern-S1	829/829	75.3	57.1	649/649	90.9	89.0	90.7	0.0
Intern-S1-Pro	824/824	31.7	11.2	694/694	57.6	33.1	51.4	2.0
Qwen3-VL-8B-Thinking	820/820	99.0	74.9	647/647	97.7	95.9	95.4	0.0
GLM-5V-Turbo	806/806	97.0	74.6	696/696	99.3	98.4	98.5	2.6
Gemini 2.5 Flash	769/769	99.1	74.3	787/787	99.6	95.1	98.9	2.2
GLM-4.6V	763/763	56.0	40.2	648/648	74.7	74.8	74.7	0.0
Kimi K2.5	730/730	99.5	78.5	636/636	100.0	100.0	98.0	0.2
Qwen3-VL-32B-Thinking	722/722	98.5	79.6	690/690	98.6	96.6	96.3	4.3
Intern-S1-Mini	661/661	48.1	23.7	650/650	77.2	66.0	76.2	0.0

Rates are percentages; fractions are counts. Item totals differ by scene: $8/16$ step relations, $4/8$ cycle relations, and $1/2$ lags in judgment/choice. Scene 2 includes the stipulated cycle-1 reference. Item accuracy is diagnostic and is not an equal-weight average of scenes. Class-adjusted recalls retain all original class episodes, including initially incorrect answers.

Table 20. **Scene-specific exact-audit scores.** Each scene–format cell pools three difficulties (432 ordinary episodes). APR conditions on the initially correct subset; LAA retains the complete ordinary denominator. All entries are percentages.

	Scene 1				Scene 2				Scene 3			
	Choice		Judgment		Choice		Judgment		Choice		Judgment	
Model	APR	LAA	APR	LAA	APR	LAA	APR	LAA	APR	LAA	APR	LAA
GPT-5.6-Sol	25.4	22.2	100.0	50.0	91.2	84.0	99.1	49.5	99.3	98.8	100.0	50.7
Claude Opus 5	15.5	12.7	85.4	50.0	79.0	66.9	99.1	52.5	93.5	92.6	94.2	85.9
Claude Sonnet 5	9.1	5.1	99.1	49.5	29.5	15.5	100.0	50.0	51.9	44.9	99.1	51.2
Gemini 2.5 Pro	4.8	2.5	71.2	47.0	39.6	22.5	86.0	68.3	20.1	17.1	61.0	58.6
GPT-4o	26.9	14.8	98.6	49.3	3.3	1.6	99.5	49.8	6.3	5.6	99.5	50.0
Intern-S1	21.7	10.9	90.7	45.4	3.3	1.6	81.9	41.0	2.0	1.9	99.1	49.8
Intern-S1-Pro	2.1	1.2	20.0	10.0	1.4	0.7	38.9	19.4	7.7	6.5	83.3	50.7
Qwen3-VL-8B-Thinking	20.9	11.3	90.3	45.4	0.0	0.0	98.6	49.3	3.5	3.0	97.7	48.4
GLM-5V-Turbo	16.5	8.6	97.2	48.4	6.9	3.2	99.5	49.8	17.6	15.5	87.2	53.5
Gemini 2.5 Flash	18.2	9.5	99.1	50.0	4.5	2.1	99.1	49.3	0.6	0.5	63.8	52.3
GLM-4.6V	14.8	8.3	94.0	47.0	1.9	0.9	30.1	15.0	6.5	4.6	100.0	50.0
Kimi K2.5	25.9	13.4	100.0	49.8	3.3	1.9	100.0	48.1	22.6	13.9	100.0	49.3
Qwen3-VL-32B-Thinking	14.9	6.7	95.8	47.7	21.6	10.6	97.2	48.6	55.6	40.5	91.1	54.6
Intern-S1-Mini	5.6	2.8	51.6	25.9	0.0	0.0	82.0	41.2	10.8	6.0	94.4	47.2

Parsing and exact versus component scoring. The parser extracts the first decodable JSON object, including from a code fence or surrounding prose. Required audit type, integer candidate IDs, candidate uniqueness and coverage, relation-list length, and allowed values are checked. Relation strings are case-insensitive; lags must be nonnegative integers, with booleans rejected. Extra JSON fields and surrounding text are not themselves penalized. An exact pass requires a valid required structure and every expected item correct. Missing or malformed required entries fail the exact score. A valid structure with even one wrong relation or lag also fails. No language-model judge or free-form explanation rating is used.

Component accuracy divides the number of correct required items by the expected item count over completed audits. Missing and invalid items receive no credit; a malformed structure can still receive credit for other correctly reported required items. Without a decodable object, every item is incorrect. Table 19 reports these micro rates; they weight Scene 1 more heavily than Scene 3 and include Scene 2's specified reference values. They therefore do not replace the scene-specific exact scores in Table 20. There is no shared 50% chance reference for these heterogeneous exact-audit tasks.

Of the 21,765 audits, 11,631 pass exactly, 7,873 have valid structures but incorrect content, and 2,261 have invalid structures. There are 730 responses with a length finish reason; this flag is reported separately and overlaps the scor-

ing categories. For example, all GPT-5.6-Sol and GPT-4o audits are structurally valid, while $563/824$ Intern-S1-Pro choice audits are invalid. This distinguishes failures to supply the requested structure from demonstrably wrong items in a valid reconstruction.

Class-conditioned interpretation. Define audit-adjusted recall for class y by

$$R_y^{\text{audit}} = \frac{\#\{\text{class } y,\ \text{answer correct, audit exact}\}}{\#\{\text{all judgment episodes of class } y\}}. \quad (11)$$

With 648 self and 648 non-self episodes, judgment LAA is the mean of these two recalls. Unlike APR, neither recall discards initially incorrect judgments. GPT-5.6-Sol's 99.7% APR is based on 648 correct self judgments and only 3 correct non-self judgments; the latter all pass, yet represent only 0.5% of the non-self population. For Gemini 2.5 Pro, $538/545$ correct self judgments and $213/498$ correct non-self judgments pass, giving class-adjusted recalls of 83.0% and 32.9%. This decomposition prevents an easy, self-heavy correct subset from being mistaken for broad contingency discrimination.

A correct choice with an incomplete reconstruction. In Scene 1, level 3, episode index 00000 (signature `e1152e1440ba`), GPT-5.6-Sol correctly chooses candidate 1. Its complete audit has the valid step-match structure, correctly reports all eight relations for candidate 1, and reports candidate 2 as shown below. Here M denotes match and X mismatch:

Step	1	2	3	4	5	6	7	8
Arm 1, expected	M	M	M	M	M	M	M	M
Arm 1, reported	M	M	M	M	M	M	M	M
Arm 2, expected	M	M	X	X	X	M	X	M
Arm 2, reported	M	M	X	X	M	M	M	M

The two missed mismatches produce $14/16$ correct items but an exact failure. The original choice remains correct: partial mismatch evidence can be sufficient for selecting the controlled arm. This example illustrates the additional reconstruction demand, not a finding that the original decision used no valid evidence. Since initially incorrect answers were not audited, their two possible audit outcomes are unobserved; a four-way answer–audit confusion matrix cannot be inferred from this protocol.

H. Controls and ablations

[Draft note: Report command-only, vision-only, final-frame-only, shuffled-command, shuffled-visual-order, and no-signed-change controls. Add simultaneous versus sequential choice and neutral "controlled/uncontrolled" wording to separate comparative evidence from self-label response bias. Include length-matched controls for the 8-step and 16-step scenes.]

I. Qualitative paired cases

Appendix G gives a verified correct-choice/failed-audit example with every required relation reported. Here we add a matched cross-format comparison in Scene 3 at level 3, base pair `00002`. Choice episode index `00005` has signature `1f3500b07be1` and self in position 2; self judgment index `00004` has signature `10d9a6955498`; non-self judgment index `00005` has signature `2300de588539`. Judgment options are 1=yes/self and 2=no/non-self for both episodes. The exact IDs, commands, provenance, and complete responses are saved in `qualitative_cases.json`.

GPT-5.6-Sol correctly chooses arm 2 and accepts the isolated self arm, but also accepts the one-step-lagged non-self arm: this triplet fails PTC. Claude Opus 5 answers all three correctly: its triplet passes. This pair was selected to illustrate both outcomes with compact complete responses, not as a random sample or estimate of their frequencies. Figure 11 shows corresponding evidence panels. All six responses below are reproduced in full; their explanation text is model output, not ground-truth annotation. Correct answers do not certify every descriptive claim in the explanations, which is why exact structured audits are scored separately.

For the displayed steps, command 1 is a wrist delta of $+0.935$ and command 2 an elbow delta of $+1.022$ (rounded). Self executes the current command; non-self executes zero at step 1 and the preceding wrist command at step 2. Thus the non-self sequence has a one-step lag even when its movements resemble the same command trace. These applied-command annotations are used for analysis and are not included in model input.

GPT-5.6-Sol

Choice: controlled arm is candidate 2.

> Choice: 2
>
> Thought: Candidate arm 2 consistently matches the trace: its wrist changes at Step 1, elbow configuration changes strongly at Step 2, and the combined base/shoulder/elbow/wrist commands produce corresponding reproducible motion through Steps 3–8. Candidate 1 shows mismatched or incidental motion on key diagnostic steps.

Isolated self: correct option is 1 (yes).

> Choice: 1
>
> Thought: Across all eight steps, visible changes occur at the commanded joint regions, with combined commands producing corresponding combined motion and opposite-sign commands reversing prior patterns. The consistent step-by-step temporal mapping supports self-control.

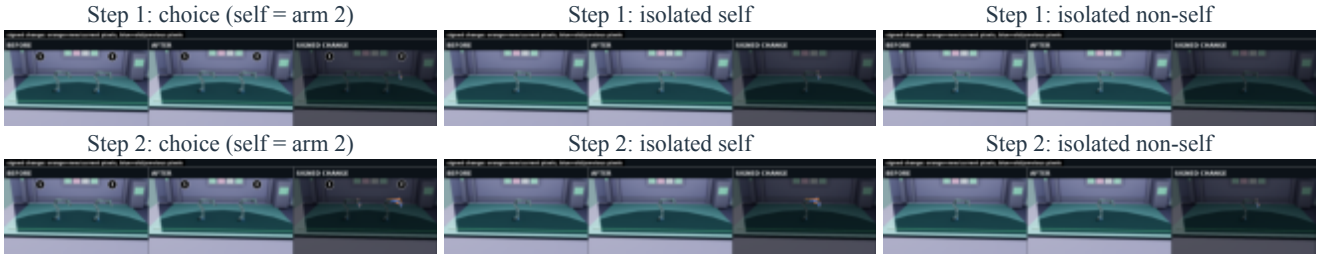


Figure 11. **Visual evidence for the matched cross-format case.** The first two of eight steps are shown for choice with self in position 2, isolated self, and isolated non-self. The underlying commands and role-aligned applied traces match across formats. Ground truth is immediate control for self and a one-step lag for non-self. Both models receive the complete eight-step evidence, not only these displayed steps.

Isolated non-self: correct option is 2 (no).

> Choice: 1
>
> Thought: The visible arm changes immediately and reproducibly with the command sequence: isolated commands affect corresponding regions, while combined commands produce combined motion, including consistent reversals in Steps 5–7. This sustained temporal correspondence supports self-control.

Claude Opus 5

Choice: controlled arm is candidate 2.

> **Choice: 2**
>
> Thought: In Step 1 (wrist-only, smallest command), the signed-change panel shows motion confined to a tiny cluster at arm 2's wrist, with arm 1 essentially unchanged — a clean one-to-one match between a small wrist command and a small end-effector-only displacement. Step 2 (elbow-only) again produces large orange/blue change across arm 2's forearm and wrist while arm 1 shows only minor incidental change. Throughout the trace, arm 2's change magnitude scales with the commanded deltas (large for the 1.13 base-yaw and 1.18 base+shoulder steps, moderate for the ˜0.86–0.98 coupled steps), and each change appears in the step immediately following the command with no lag. Arm 1 only begins moving from Step 3 onward and its motion does not track the per-joint pattern (e.g., it shows broad body motion on wrist-dominant steps and little response to the isolated wrist command), indicating it is an independently scripted distractor. The consistent, immediate, magnitude-proportional command-to-motion coupling identifies arm 2 as self-controlled.

Isolated self: correct option is 1 (yes).

> Choice: 1
>
> Thought: Across all eight steps the visible change magnitude and location track the commanded joints: Step 1 (wrist-only, small delta) produced a tiny change blob confined to the end-effector; Step 2 (elbow) produced a large change spanning the forearm/upper link; Steps 3–4 (base yaw, with/without wrist) shifted the whole upper arm laterally with the base largely fixed; Step 5 (base yaw + shoulder, largest deltas) produced the most extensive change including the proximal link; Steps 6–8 (shoulder/elbow combinations) again moved the corresponding proximal-to-distal segments with magnitudes proportional to the commanded values. The changes occur immediately after each command with no unexplained motion during zero-delta joints, indicating a consistent, predictable command-to-motion coupling characteristic of self-control.

Isolated non-self: correct option is 2 (no).

> Choice: 2
>
> Thought: The command-to-motion mapping is not consistent. Large wrist (0.94) and elbow (1.02) commands in Steps 1–2 produced essentially no visible articulation (Step 1 change image is blank; Step 2 shows only a few speckles at the tool tip, not a forearm swing). Steps 3–4 with substantial base_yaw (0.88, 1.13) should swing the entire arm laterally about the base, yet the base/lower link pixels stay fixed and only the distal links show small differences. Likewise Step 5's combined negative base_yaw+shoulder (-1.18 each) yields a smaller change footprint than Step 6's smaller (0.86) command. This lack of proportionality and of joint-specific, reproducible correspondence indicates the observed arm's motion is driven by something other than this command trace.

J. Physical-robot protocol

Recording design. The frozen `ur5_real_v1` dataset was collected on September 19, 2026. All 108 recorded episodes are retained; no experimental data were excluded. One UR5 arm is observed in a real workspace from a fixed external camera. The intended control vector has four labeled coordinates: `base_yaw`, `shoulder`, `elbow`, and `wrist`. These are the labels supplied to the model, not an inferred physical joint calibration. Each of the nine scene–difficulty cells contains six self and six non-self episodes. The same six base pair identifiers within a scene are reused across the three difficulties, giving 18 base clusters and 54 condition pairs across levels. The correct option appears equally often in positions 1 and 2, including within each class in each cell.

The three interventions retain the command-defined conditions in Table 2: same-step mismatch counts of $8/6/4$ in an eight-step sequence; four four-step cycles with non-self mapping patterns A–B–C–D, A–A–B–C, and A–A–A–B; and $3/2/1$-step delayed commands in an eight-step sequence. Self follows the current command in Scenes 1 and 3 and one fixed hidden mapping across cycles in Scene 2. Delay is measured in discrete command steps, not milliseconds. Scene 1 contributes 288 steps, Scene 2 contributes 576, and Scene 3 contributes 288.

Table 21. **Self-attribution on physical UR5 recordings.** All five models evaluate the same 108 balanced judgment episodes. S1–S3 are scene accuracies averaged over difficulty. Overall accuracy equals balanced accuracy; brackets give marginal 95% scene-stratified base-pair bootstrap intervals. Self/non-self recall use 54 episodes per class. Self attribution is the percentage of valid answers that label the arm self.

Model	S1	S2	S3	Overall [95% CI]	Self	Non-self	Self attr.
GPT-5.6-Sol	50.0	50.0	50.0	50.0 [50.0, 50.0]	100.0	0.0	100.0
Claude Opus 5	52.8	50.0	72.2	58.3 [54.6, 62.0]	100.0	16.7	91.7
Gemini 2.5 Pro	50.0	66.7	91.7	69.4 [63.0, 75.0]	87.0	51.9	67.6
Qwen3-VL-32B-Thinking	50.0	50.0	50.0	50.0 [50.0, 50.0]	100.0	0.0	100.0
GLM-5V-Turbo	50.0	50.0	55.6	51.9 [50.0, 53.7]	100.0	3.7	98.1

Table 22. **Complete physical-robot results by scene and difficulty.** Each cell contains 12 judgment episodes: six self and six non-self. Acc., S, and N denote accuracy, self recall, and non-self recall (%). Invalid outputs count as incorrect; none occurred in these 540 evaluations.

		Easy			Medium			Hard		
Model	Scene	Acc.	S	N	Acc.	S	N	Acc.	S	N
GPT-5.6-Sol	S1	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S2	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S3	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
Claude Opus 5	S1	58.3	100.0	16.7	50.0	100.0	0.0	50.0	100.0	0.0
	S2	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S3	83.3	100.0	66.7	66.7	100.0	33.3	66.7	100.0	33.3
Gemini 2.5 Pro	S1	50.0	100.0	0.0	58.3	100.0	16.7	41.7	83.3	0.0
	S2	66.7	100.0	33.3	83.3	83.3	83.3	50.0	50.0	50.0
	S3	100.0	100.0	100.0	91.7	83.3	100.0	83.3	83.3	83.3
Qwen3-VL-32B-Thinking	S1	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S2	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S3	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
GLM-5V-Turbo	S1	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S2	50.0	100.0	0.0	50.0	100.0	0.0	50.0	100.0	0.0
	S3	66.7	100.0	33.3	50.0	100.0	0.0	50.0	100.0	0.0

Evidence and ground truth. Each episode stores its initial 1280×720 camera frame. Each step stores the intended and applied control vectors, the action-success flag, start/end timestamps, a camera frame at the same resolution, and a 1536×352 evidence panel. The panel places BEFORE, AFTER, and SIGNED CHANGE views side by side; orange marks new/current pixels and blue marks old/previous pixels. The evaluation input contains all eight or sixteen ordered panels and the final full camera frame, together with the intended command trace. Applied vectors, candidate roles, mismatch counts, mapping schedules, lag values, and target flags are withheld from the model.

We verify all image and episode checksums and reconstruct each model's multimodal-input hash from the saved images and intended commands. We also check that the applied command traces realize every scene–difficulty condition. All 1,152 steps carry a successful action flag. This validates the logged command process; the flags are not a quantitative measure of physical tracking error. The frozen archive does not retain encoder-based tracking measurements, camera intrinsics or make, detailed controller configuration, or the conversion from normalized command units to physical joint displacement. Reproducing the offline analysis uses only the preserved artifacts; repeating acquisition requires those additional rig settings.

Models and inference. The physical snapshot includes every available complete run: GPT-5.6-Sol (`gpt-5.6-sol`), Claude Opus 5 (`claude-opus-5`), Gemini 2.5 Pro (`gemini-2.5-pro`), Qwen3-VL-32B-Thinking (`Qwen/Qwen3-VL-32B-Thinking`), and GLM-5V-Turbo (`glm-5v-turbo`). Evaluations were recorded on October 10–11, 2026 under `realworld_eval_v1_compatible`, with high-detail images, unrestricted saved image history, and a maximum of 8,192 completion tokens. Temperature is omitted. The real-world prompt differs from the simulation prompt in its description of the visual domain; both request the numbered choice before the explanation. No model controls a live robot. All 540 ordinary responses are valid, with one result per model and episode and no missing cells. The analysis uses the ordinary results rather than a selected subset or a

retry-based score.

Estimator and uncertainty. Overall accuracy is the equal-weight mean over the nine balanced cells, equivalent to pooling 108 episodes. Self and non-self recall pool 54 episodes of their respective classes. Scene accuracies pool 36 episodes, and difficulty summaries pool 36 episodes across scenes. Self attribution is the fraction of valid responses selecting the semantic yes/self option, accounting for counterbalanced option order. All invalid responses would remain in accuracy and recall denominators and count as incorrect.

For 5,000 replicates (seed 2026), we independently draw six base nuisance pairs with replacement within each scene. Every sampled pair contributes both answer conditions and its three recorded difficulty levels. The same draws are shared across models, difficulties, and metrics. We recompute the relevant means and take the 2.5th and 97.5th percentiles. These pointwise intervals quantify variation among the recorded base pairs; they do not cover changes of robot, workspace, camera, or provider. The two all-self models have $[50.0, 50.0]\%$ accuracy intervals because each balanced pair contains one accepted self and one falsely accepted non-self. Such a degenerate resampling interval describes their constant policy on this dataset, not certainty about deployment performance.

Scope and reproducibility. The physical experiment contains judgment only. It establishes that strong self acceptance and weak non-self rejection can persist in physical recordings; it cannot estimate a physical choice advantage or PTC. The single rig and small number of base pairs also preclude attributing the simulation/physical score difference solely to rendering quality. `summarize_realworld_paper.py` validates coverage, labels, source records, command-defined interventions, and the shared inputs, then generates Tables 21 and 22 from the frozen result tree. The dataset content hash begins `5885bd9d5fcd5fe8`; the analysis JSON records the full hash, every source-result checksum, requested model identifiers, evaluation dates, and bootstrap configuration.

K. Reproducibility artifacts

[Draft note: List the released code, frozen manifests, content hashes, simulator version, assets and licenses, random seeds, raw and parsed model outputs, table generation scripts, and instructions for reproducing every figure and table.]

L. Benchmark comparison and coverage ratings

Table 1 combines quantities reported by the cited benchmarks with two author-assessed ordinal design-coverage ratings. The stars are neither reported benchmark scores nor estimates of model performance. One star means that the relevant diagnostic target is not explicitly evaluated, not that the benchmark is of low overall quality.

Quantities and units. Questions, instructions, test tasks, task categories, episodes, and interaction steps retain their original units and must not be ranked as interchangeable sample counts. VLABench's 100 counts task categories, whereas MirrorBench's more than 5,000 counts inference steps per model; the latter is not used to infer a fixed episode count. MVBench's 20 temporal tasks and TempCompass's 5 temporal aspects (10 subtypes) are not counted as motor-control interventions. NavBench has 1,200 global-alignment, 1,000 progress-estimation, and 500 examples for each local reasoning variant, giving 3,200 comprehension questions, separately from 432 navigation cases. Its three task-difficulty bins, MirrorBench's four prompt-prior conditions, and VISUOSELF's three intervention-separability levels describe different graded factors. "–" denotes no reported uniform fixed levels, not an absence of challenging examples. Counts in parentheses identify ordinary answer formats where comparable; reconstruction audits are not counted as a third ordinary format for VISUOSELF.

Bodily self-recognition. The five levels are: (1) no explicit bodily self-recognition task; (2) indirect use of information about the agent's body without a direct self-identification test; (3) an explicit but isolated self-identification task; (4) direct testing of bodily correspondence or self/other distinction; and (5) a dedicated controlled, multi-condition self-recognition protocol. MirrorBench and VISUOSELF both receive five stars. Their targets differ: MirrorBench uses the mirror-mark paradigm with graded prior information, whereas VISUOSELF uses command–motion relations with graded non-self interventions. The other selected benchmarks primarily evaluate external perception or task objectives; their one-star entries refer only to this specific diagnostic target.

Cross-format diagnosis. The five levels are: (1) no dedicated cross-format diagnostic analysis; (2) comparison of complementary decision protocols in different tasks; (3) multiple formats grounded in the same underlying content; (4) systematic performance comparisons across such content-linked formats; and (5) a joint-correctness criterion over explicitly matched responses. Formats here concern

decision protocols rather than output syntax alone. TempCompass receives four stars for its four content-linked formats and their systematic comparison. NavBench receives two stars for analyzing comprehension and execution; this is not treated as matched cross-format joint correctness. VISUOSELF receives five stars for PTC and PTC_4 on 648 matched four-response sets, with 216 persistent base pairs across difficulty levels. The $1{,}296$ PTC triplets reuse judgment responses and are not independent observations.

Process truth and validation scope. The last column names actual annotation content or scoring evidence rather than assigning a generic annotation-quality score. Stepwise actions and states in VLABench and hand–mark distance trajectories in MirrorBench remain process-level evidence. VISUOSELF additionally scores structured stepwise match/mismatch, cross-cycle mapping, and exact-lag reconstructions against applied command traces. Its audits cover initially correct simulation answers; they do not establish the internal mechanism producing those answers. The separate 108 physical episodes contain judgment only and cannot establish physical choice performance or PTC.